# Unit 1: Numerical Expressions & Factors — Practice A & B Complete Key

**Big Ideas Math Grade 6 — Chapter 1 • Manipulative Kits: Counting & Array Studio (COUNT-01) + Prime Factor Studio (PRIME-01)**

**Target Misconception: Students interpret remainders as meaningless decimal tails (e.g., .875 of a person), read 2^3 as 2×3=6, lose track of PEMDAS parentheses, and confuse GCF (overlap) with LCM (union).**

Hands-On Block Remediation: Have students build physical tile rows for division remainders, stack 2D/3D arrays for exponents, use the 4-line elevator (Y holds Room 1 while X solves Room 2) instead of rewriting PEMDAS lines, and stand logarithmic Prime Factor Towers side-by-side on the 12-socket lid.



`QUEST-01 Double-Points Scoring: 1 '+10' token = Textbook Method 1 (1st Way); stack a 2nd '+10' token (+20 pts) = Textbook Method 2 (2nd Way)!`

## SEC 1.1 (pp.4-5): WHOLE NUMBER OPERATIONS & DIVISION WITH REMAINDERS [COUNT-01]

Methods: W1: Standard Long Division | W2: Partial Quotients | W3: Equal-Groups Array Model

### PRACTICE A (p.4): KEY IDEAS (#1-#10) — ExA: 5 R 3

**A1: 5 R 1**
Setup: 31 ÷ 6 (Q & R)
SC-6: 31 [ENTER] 6 [g][÷R]

**A2: 9 R 2**
Setup: 47 ÷ 5 (Q & R)
SC-6: 47 [ENTER] 5 [g][÷R]

**A3: 8 R 2**
Setup: 58 ÷ 7 (Q & R)
SC-6: 58 [ENTER] 7 [g][÷R]

**A4: 9 R 4**
Setup: 85 ÷ 9 (Q & R)
SC-6: 85 [ENTER] 9 [g][÷R]

**A5: 12 R 2**
Setup: 146 ÷ 12 (Q & R)
SC-6: 146 [ENTER] 12 [g][÷R]

**A6: 7 full lines, 5 cm left**
Setup: 250 ÷ 35 (Q & R)
SC-6: 250 [ENTER] 35 [g][÷R]

**A7: 117**
Setup: (14 × 8) + 5
SC-6: 14 [ENTER] 8 [×] 5 [+]

**A8: 9 R 3 (Rem < 8)**
Setup: 75 ÷ 8 (Q & R)
SC-6: 75 [ENTER] 8 [g][÷R]

**A9: 19 rovers, 2 tires left**
Setup: 78 ÷ 4 (Q & R)
SC-6: 78 [ENTER] 4 [g][÷R]

**A10: 8 tents (7 full + 1 for the 29th explorer)**
Setup: 29 ÷ 4 (Q & R)
SC-6: 29 [ENTER] 4 [g][÷R]

### PRACTICE B (p.5): REAL LIFE (#1-#10) — ExB: 17 vans (16 full vans + 1 van with 7 students)

**B1: 8 full packs, 5 cells left**
Setup: 53 ÷ 6 (Q & R)
SC-6: 53 [ENTER] 6 [g][÷R]

**B2: 11 figures, 6 tickets left**
Setup: 94 ÷ 8 (Q & R)
SC-6: 94 [ENTER] 8 [g][÷R]

**B3: 20 canoes (19 full + 1 with 2 campers)**
Setup: 78 ÷ 4 (Q & R)
SC-6: 78 [ENTER] 4 [g][÷R]

**B4: 9 full racks, 2 on last**
Setup: 110 ÷ 12 (Q & R)
SC-6: 110 [ENTER] 12 [g][÷R]

**B5: 28 knights, 2 m left**
Setup: 450 ÷ 16 (Q & R)
SC-6: 450 [ENTER] 16 [g][÷R]

**B6: 18 full, 19 caches total**
Setup: 165 ÷ 9 (Q & R)
SC-6: 165 [ENTER] 9 [g][÷R]

**B7: 187 marbles**
Setup: (12 × 15) + 7
SC-6: 12 [ENTER] 15 [×] 7 [+]

**B8: 8 full trains, 8 on final**
Setup: 200 ÷ 24 (Q & R)
SC-6: 200 [ENTER] 24 [g][÷R]

**B9: 9 pizzas (5 slices left)**
Setup: 67 ÷ 8 (Q & R)
SC-6: 67 [ENTER] 8 [g][÷R]

**B10: 11 full crates, 17 apples left**
Setup: 325 ÷ 28 (Q & R)
SC-6: 325 [ENTER] 28 [g][÷R]

## SEC 1.2 (pp.6-7): POWERS AND EXPONENTS [COUNT-01]

Methods: W1: Multiply Step-by-Step | W2: Group Factors in Pairs | W3: Square & Cube Array Model

### PRACTICE A (p.6): KEY IDEAS (#1-#10) — ExA: 4³ = 16, 4⁴ = 64

**A1: 5 × 5 = 25**
Setup: 5²
SC-6: 5 [x²]

**A2: 49 (vs. 14)**
Setup: 7²
SC-6: 7 [x²]

**A3: 3 × 3 × 3 = 27**
Setup: 3³
SC-6: 3 [x³]

**A4: 125**
Setup: 5³
SC-6: 5 [x³]

**A5: 32**
Setup: 2^5
SC-6: 2 [ENTER] 5 [g][y^x]

**A6: 1,296**
Setup: 6^4
SC-6: 6 [ENTER] 4 [g][y^x]

**A7: 10,000 (4 zeros) and 1**
Setup: 10^4
SC-6: 10 [ENTER] 4 [g][y^x]

**A8: 16 (2×2×2×2 = 16)**
Setup: 2^4
SC-6: 2 [ENTER] 4 [g][y^x]

**A9: 96 kN (3 × 32)**
Setup: 2^5 × 3
SC-6: 2 [ENTER] 5 [g][y^x] 3 [×]

**A10: 8 m³ per room; 64 m³ total**
Setup: 2³ × 8
SC-6: 2 [x³] 8 [×]

### PRACTICE B (p.7): REAL LIFE (#1-#10) — ExB: 81 shares

**B1: 8² = 64 pixels**
Setup: 8²
SC-6: 8 [x²]

**B2: 6³ = 216 blocks**
Setup: 6³
SC-6: 6 [x³]

**B3: 2^6 = 64 bacteria**
Setup: 2^6
SC-6: 2 [ENTER] 6 [g][y^x]

**B4: 4 × 4 × 4 × 4 = 256 players**
Setup: 4^4
SC-6: 4 [ENTER] 4 [g][y^x]

**B5: 144 + 25 = 169 ft²**
Setup: 12² + 5²
SC-6: 12 [x²] 5 [x²] [+]

**B6: 5^4 = 625 starlings**
Setup: 5^4
SC-6: 5 [ENTER] 4 [g][y^x]

**B7: 14 meters**
Setup: √(196)
SC-6: 196 [f][√x]

**B8: 2^5 = 32 > 5² = 25 (by 7)**
Setup: 2^5 - 5²
SC-6: 2 [ENTER] 5 [g][y^x] 5 [x²] [-]

**B9: 1,024 bytes**
Setup: 2^10
SC-6: 2 [ENTER] 10 [g][y^x]

**B10: 27 + 343 = 370 in³**
Setup: 3³ + 7³
SC-6: 3 [x³] 7 [x³] [+]

## SEC 1.3 (pp.8-9): ORDER OF OPERATIONS [COUNT-01]

Methods: W1: PEMDAS Step-by-Step Funnel | W2: Distributive Property / Fraction-Bar Grouping

### PRACTICE A (p.8): KEY IDEAS (#1-#10) — ExA: 72

**A1: 11**
Setup: 5 + (2 × 3)
SC-6: 5 [ENTER] 2 [ENTER] 3 [×] [+]

**A2: 4**
Setup: (14 + 6) ÷ (7 - 2)
SC-6: 14 [ENTER] 6 [+] 7 [ENTER] 2 [-] [÷]

**A3: 6**
Setup: 30 - (4 × 6)
SC-6: 30 [ENTER] 4 [ENTER] 6 [×] [-]

**A4: 84**
Setup: (9 + 3) × (15 - 8)
SC-6: 9 [ENTER] 3 [+] 15 [ENTER] 8 [-] [×]

**A5: 1**
Setup: 50 - (3 + 4)²
SC-6: 50 [ENTER] 3 [ENTER] 4 [+] [x²] [-]

**A6: 20**
Setup: 120 - (4 × (3 + 2)²)
SC-6: 120 [ENTER] 4 [ENTER] 3 [ENTER] 2 [+] [x²] [×] [-]

**A7: 2 (24 ÷ 12)**
Setup: (48 - (4 × 6)) ÷ (3 + (2 × 4.5))
SC-6: 48 [ENTER] 4 [ENTER] 6 [×] [-] 3 [ENTER] 2 [ENTER] 4.5 [×] [+]..

**A8: 24**
Setup: (3³ + (15 ÷ 3)) - (2 × 4)
SC-6: 3 [x³] 15 [ENTER] 3 [÷] [+] 2 [ENTER] 4 [×] [-]

**A9: $92 (60 + 32)**
Setup: (5 × 12) + (4 × 8)
SC-6: 5 [ENTER] 12 [×] 4 [ENTER] 8 [×] [+]

**A10: $62 (90 - 28)**
Setup: (45 × 2) - 28
SC-6: 45 [ENTER] 2 [×] 28 [-]

### PRACTICE B (p.9): REAL LIFE (#1-#10) — ExB: 25

**B1: 4(9 + 5) = $56**
Setup: (9 + 5) × 4
SC-6: 9 [ENTER] 5 [+] 4 [×]

**B2: 45 + 4(7) = $73**
Setup: 45 + (4 × 7)
SC-6: 45 [ENTER] 4 [ENTER] 7 [×] [+]

**B3: 101 points**
Setup: (100 - (3 × 8)) ÷ 25
SC-6: 100 [ENTER] 3 [ENTER] 8 [×] [-] 25 [÷]

**B4: 10³ - 6³ = 64 m³**
Setup: 10³ - 6³
SC-6: 10 [x³] 6 [x³] [-]

**B5: $73**
Setup: (2 × 14) + (5 × 9)
SC-6: 2 [ENTER] 14 [×] 5 [ENTER] 9 [×] [+]

**B6: 70 points**
Setup: ((3² + 5) × 6) - 14
SC-6: 3 [x²] 5 [+] 6 [×] 14 [-]

**B7: 10 bracelets per box**
Setup: (72 - 12) ÷ 6
SC-6: 72 [ENTER] 12 [-] 6 [÷]

**B8: 42 (÷ and × go left to right)**
Setup: ((48 ÷ 6) × 4) + 10
SC-6: 48 [ENTER] 6 [÷] 4 [×] 10 [+]

**B9: 12 muffins per basket**
Setup: ((5 × 12) + (3 × 8)) ÷ 7
SC-6: 5 [ENTER] 12 [×] 3 [ENTER] 8 [×] [+] 7 [÷]

**B10: 74**
Setup: ((4³ - 14) ÷ (2 + 3)) + 8²
SC-6: 4 [x³] 14 [-] 2 [ENTER] 3 [+] [÷] 8 [x²] [+]

## SEC 1.4 (pp.10-11): PRIME FACTORIZATION [PRIME-01]

Methods: W1: Factor Tree (Pair A) | W2: Factor Tree (Pair B) | W3: Division Ladder

### PRACTICE A (p.10): KEY IDEAS (#1-#10) — ExA: 60 = 2² × 3 × 5

**A1: 2² × 3**
Setup: Factors(12)
SC-6: 12 [g][FACT]

**A2: 2² × 3²**
Setup: Factors(36)
SC-6: 36 [g][FACT]

**A3: 29 is prime; 91 is composite (7 × 13)**
Setup: Factors(91)
SC-6: 29 [PRIME], 91 [g][FACT]

**A4: 3² × 5**
Setup: Factors(45)
SC-6: 45 [g][FACT]

**A5: 2³ × 3²**
Setup: Factors(72)
SC-6: 72 [g][FACT]

**A6: 2² × 3 × 7**
Setup: Factors(84)
SC-6: 84 [g][FACT]

**A7: 2 × 3 × 5 × 7**
Setup: Factors(210)
SC-6: 210 [g][FACT]

**A8: 2³ × 3² × 7**
Setup: Factors(504)
SC-6: 504 [g][FACT]

**A9: 7 × 11 × 13**
Setup: Factors(1001)
SC-6: 1001 [g][FACT]

**A10: 2^4 × 3² × 5²**
Setup: Factors(3600)
SC-6: 3600 [g][FACT]

### PRACTICE B (p.11): REAL LIFE (#1-#10) — ExB: 2² × 3 × 5 (e.g., 8 × 15 or 10 × 12)

**B1: 1×48, 2×24, 3×16, 4×12, 6×8; 2^4 × 3**
Setup: Factors(48)
SC-6: 48 [g][FACT]

**B2: 2 × 3³**
Setup: Factors(54)
SC-6: 54 [g][FACT]

**B3: 51 is composite (3 × 17)**
Setup: Factors(51)
SC-6: 51 [g][FACT]

**B4: 2^4 × 5**
Setup: Factors(80)
SC-6: 80 [g][FACT]

**B5: 126**
Setup: (2 × 3²) × 7
SC-6: 2 [ENTER] 3 [x²] [×] 7 [×]

**B6: 2² × 5²**
Setup: Factors(100)
SC-6: 100 [g][FACT]

**B7: 2² × 5 × 7**
Setup: Factors(140)
SC-6: 140 [g][FACT]

**B8: 2^4 × 3²**
Setup: Factors(144)
SC-6: 144 [g][FACT]

**B9: Both give 2 × 3² × 5**
Setup: Factors(90)
SC-6: 90 [g][FACT]

**B10: 2 × 3 × 5 × 7 = 210**
Setup: ((2 × 3) × 5) × 7
SC-6: 2 [ENTER] 3 [×] 5 [×] 7 [×]

## SEC 1.5 (pp.12-13): GREATEST COMMON FACTOR (GCF) & LEAST COMMON MULTIPLE (LCM) [PRIME-01]

Methods: W1: List Factors & Multiples | W2: Prime Factorization | W3: Prime Factor Venn Diagram

### PRACTICE A (p.12): KEY IDEAS (#1-#10) — ExA: GCF = 10, LCM = 100

**A1: GCF = 6, LCM = 36**
Setup: GCF(12, 18); LCM(12, 18)
SC-6: 12 [ENTER] 18 [GCF] / [f][LCM]

**A2: GCF = 4, LCM = 24**
Setup: GCF(8, 12); LCM(8, 12)
SC-6: 8 [ENTER] 12 [GCF] / [f][LCM]

**A3: GCF = 5, LCM = 75**
Setup: GCF(15, 25); LCM(15, 25)
SC-6: 15 [ENTER] 25 [GCF] / [f][LCM]

**A4: GCF = 7, LCM = 70**
Setup: GCF(14, 35); LCM(14, 35)
SC-6: 14 [ENTER] 35 [GCF] / [f][LCM]

**A5: 24**
Setup: GCF(48, 72)
SC-6: 48 [ENTER] 72 [GCF]

**A6: 28**
Setup: GCF(84, 140)
SC-6: 84 [ENTER] 140 [GCF]

**A7: GCF = 15, LCM = 225**
Setup: GCF(45, 75); LCM(45, 75)
SC-6: 45 [ENTER] 75 [GCF] / [f][LCM]

**A8: 60 minutes**
Setup: LCM(15, 20)
SC-6: 15 [ENTER] 20 [f][LCM]

**A9: 180 seconds (3 minutes)**
Setup: LCM(45, 60)
SC-6: 45 [ENTER] 60 [f][LCM]

**A10: 12 bags (3 red, 4 blue per bag)**
Setup: GCF(36, 48)
SC-6: 36 [ENTER] 48 [GCF]

### PRACTICE B (p.13): REAL LIFE (#1-#10) — ExB: 6 packs (3 pretzels and 4 fruit bars per pack)

**B1: 14 kits (2 bandages, 3 wipes each)**
Setup: GCF(28, 42)
SC-6: 28 [ENTER] 42 [GCF]

**B2: 60 years**
Setup: LCM(12, 15)
SC-6: 12 [ENTER] 15 [f][LCM]

**B3: 24 (3 packs of hot dogs, 2 packs of buns)**
Setup: LCM(8, 12)
SC-6: 8 [ENTER] 12 [f][LCM]

**B4: 16 singers per row**
Setup: GCF(32, 48)
SC-6: 32 [ENTER] 48 [GCF]

**B5: 10:12 AM (72 minutes later)**
Setup: LCM(18, 24)
SC-6: 18 [ENTER] 24 [f][LCM]

**B6: 12 inches**
Setup: GCF(36, 60)
SC-6: 36 [ENTER] 60 [GCF]

**B7: GCF = 7, LCM = 42**
Setup: LCM(14, 21)
SC-6: 14 [ENTER] 21 [GCF] 14 [ENTER] 21 [f][LCM]

**B8: 60 minutes**
Setup: LCM(LCM(4, 6), 10)
SC-6: 4 [ENTER] 6 [f][LCM] 10 [f][LCM]

**B9: 15 bouquets (3 gold, 5 silver each)**
Setup: GCF(45, 75)
SC-6: 45 [ENTER] 75 [GCF]

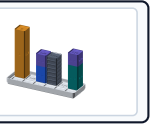
**B10: GCF=5, LCM=60; 5×60 = 300 = 15×20**
Setup: GCF(15, 20) × LCM(15, 20)
SC-6: 15 [ENTER] 20 [GCF] 15 [ENTER] 20 [f][LCM] [×]

# Unit 2: Fractions & Decimals — Practice A & B Complete Key

**Big Ideas Math Grade 6 — Chapter 2 • Manipulative Kits: Fraction Studio (FRAC-01) + Place Value Studio (PLACE-01)**

**Target Misconception: Students treat fractions, mixed numbers, and decimals as unrelated rules, expect multiplication to always make numbers bigger, forget which fraction to flip in division, and misalign decimal columns.**

Hands-On Block Remediation: Use the 84.0 mm Fraction Studio towers so equivalence and common denominators are physical millimeter height matches. Use [f] [1/x] on Floor 1 (X) so students see only the divisor flip, and use [f] [IP] / [g] [FP] to split whole dollars from decimal cents.



**QUEST-01 Double-Points Scoring: 1 '+10' token = Textbook Method 1 (1st Way); stack a 2nd '+10' token (+20 pts) = Textbook Method 2 (2nd Way)!**

## SEC 2.1 (pp.15-16): EQUIVALENT FRACTIONS, MIXED NUMBERS & LCD ADDITION [FRAC-01]

Methods: W1: Divide Once by GCF / Use LCD | W2: Step-by-Step Common Factors | W3: Fraction Strips Model

### PRACTICE A (p.15): KEY IDEAS (#1-#10) — ExA: 2/3

**A1: 2/3**
Setup: Simp(8/12)×2
SC-6: [.] 8 [.] 12 [ENTER] [g][SIMP]×2

**A2: 8/3 = 2 2/3 trays**
Setup: Simp(48/18)
SC-6: [.] 48 [.] 18 [ENTER] [g][SIMP]

**A3: 13/5 and 2.6**
Setup: Improper(2 3/5)
SC-6: 2 [.] 3 [.] 5 [ENTER] [f][d/c]

**A4: 3 5/6 pizzas (3 whole, 5 sixths)**
Setup: 23/6
SC-6: [.] 23 [.] 6 [ENTER]

**A5: 5/6**
Setup: 1/2 + 1/3
SC-6: [.] 1 [.] 2 [ENTER] [.] 1 [.] 3 [+]

**A6: 1 3/20 gallons (23/20)**
Setup: 3/4 + 2/5
SC-6: [.] 3 [.] 4 [ENTER] [.] 2 [.] 5 [+]

**A7: 3 5/6 feet**
Setup: 1 1/2 + 2 1/3
SC-6: 1 [.] 1 [.] 2 [ENTER] 2 [.] 1 [.] 3 [+]

**A8: 2 7/12 feet**
Setup: 5 1/3 - 2 3/4
SC-6: 5 [.] 1 [.] 3 [ENTER] 2 [.] 3 [.] 4 [-]

**A9: 7/8 climbed; 1/8 remains**
Setup: (1/2 + 1/4) + 1/8
SC-6: [.] 1 [.] 2 [ENTER] [.] 1 [.] 4 [+] [.] 1 [.] 8 [+]

**A10: 5 1/4 pints**
Setup: 3 1/2 + 1 3/4
SC-6: 3 [.] 1 [.] 2 [ENTER] 1 [.] 3 [.] 4 [+]

### PRACTICE B (p.16): REAL LIFE (#1-#10) — ExB: 2 11/12 miles (35/12 miles)

**B1: 5/8 cup**
Setup: 3/8 + 1/4
SC-6: [.] 3 [.] 8 [ENTER] [.] 1 [.] 4 [+]

**B2: 3/4 complete; 1/4 remains**
Setup: Simp(18/24)
SC-6: [.] 18 [.] 24 [g][SIMP]

**B3: 13/8 and 1.625 in**
Setup: Improper(1 5/8)
SC-6: 1 [.] 5 [.] 8 [ENTER] [f][d/c] [FDISP]

**B4: 11/24 foot**
Setup: 5/6 - 3/8
SC-6: [.] 5 [.] 6 [ENTER] [.] 3 [.] 8 [-]

**B5: 3 5/6 pizzas (23/6)**
Setup: 2 1/3 + 1 1/2
SC-6: 2 [.] 1 [.] 3 [ENTER] 1 [.] 1 [.] 2 [+]

**B6: 2 3/4 feet (11/4)**
Setup: 4 1/2 - 1 3/4
SC-6: 4 [.] 1 [.] 2 [ENTER] 1 [.] 3 [.] 4 [-]

**B7: 3/5 = 0.6**
Setup: Simp(15/25)
SC-6: [.] 15 [.] 25 [g][SIMP] [FDISP]

**B8: 1/5 liter (0.2 L)**
Setup: (2/5 + 3/10) - 1/2
SC-6: [.] 2 [.] 5 [ENTER] [.] 3 [.] 10 [+] [.] 1 [.] 2 [-]

**B9: 4 5/12 yd used; 7/12 yd left**
Setup: 2 2/3 + 1 3/4
SC-6: 2 [.] 2 [.] 3 [ENTER] 1 [.] 3 [.] 4 [+]

**B10: 4 5/8 miles**
Setup: (3/8 + 2 1/2) + 1 3/4
SC-6: [.] 3 [.] 8 [ENTER] 2 [.] 1 [.] 2 [+] 1 [.] 3 [.] 4 [+]

## SEC 2.2 (pp.17-18): MULTIPLYING FRACTIONS AND MIXED NUMBERS [FRAC-01]

Methods: W1: Multiply Straight Across | W2: Cross-Cancel Common Factors First | W3: Area Model / Distributive Split

### PRACTICE A (p.17): KEY IDEAS (#1-#10) — ExA: 3/8

**A1: 1/3**
Setup: 1/2 × 2/3
SC-6: [.] 1 [.] 2 [ENTER] [.] 2 [.] 3 [×]

**A2: 1/2**
Setup: 2/3 × 3/4
SC-6: [.] 2 [.] 3 [ENTER] [.] 3 [.] 4 [×]

**A3: 1/3**
Setup: 3/5 × 5/9
SC-6: [.] 3 [.] 5 [ENTER] [.] 5 [.] 9 [×]

**A4: 6 cups**
Setup: 16 × 3/8
SC-6: 16 [ENTER] [.] 3 [.] 8 [×]

**A5: 4**
Setup: 1 3/4 × 2 2/7
SC-6: 1 [.] 3 [.] 4 [ENTER] 2 [.] 2 [.] 7 [×]

**A6: 3**
Setup: 2 1/4 × 1 1/3
SC-6: 2 [.] 1 [.] 4 [ENTER] 1 [.] 1 [.] 3 [×]

**A7: 9 cups**
Setup: 2 1/4 × 4
SC-6: 2 [.] 1 [.] 4 [ENTER] 4 [×]

**A8: 3 1/3 (Less than 5 because 2/3 < 1)**
Setup: 5 × 2/3
SC-6: 5 [ENTER] [.] 2 [.] 3 [×]

**A9: 14 7/8 miles (119/8)**
Setup: 3 1/2 × 4 1/4
SC-6: 3 [.] 1 [.] 2 [ENTER] 4 [.] 1 [.] 4 [×]

**A10: 1/4 of the input speed**
Setup: (1/2 × 2/3) × 3/4
SC-6: [.] 1 [.] 2 [ENTER] [.] 2 [.] 3 [×] [.] 3 [.] 4 [×]

### PRACTICE B (p.18): REAL LIFE (#1-#10) — ExB: 4 cups

**B1: 2/5 of the garden**
Setup: 2/3 × 3/5
SC-6: [.] 2 [.] 3 [ENTER] [.] 3 [.] 5 [×]

**B2: 2/5 of a full battery**
Setup: 4/5 × 1/2
SC-6: [.] 4 [.] 5 [ENTER] [.] 1 [.] 2 [×]

**B3: 4 3/8 cups (35/8)**
Setup: 1 3/4 × 2 1/2
SC-6: 1 [.] 3 [.] 4 [ENTER] 2 [.] 1 [.] 2 [×]

**B4: 3 in²**
Setup: 2 1/4 × 1 1/3
SC-6: 2 [.] 1 [.] 4 [ENTER] 1 [.] 1 [.] 3 [×]

**B5: 4 miles**
Setup: 3 1/5 × 1 1/4
SC-6: 3 [.] 1 [.] 5 [ENTER] 1 [.] 1 [.] 4 [×]

**B6: 2/3 yd²**
Setup: 3/4 × 8/9
SC-6: [.] 3 [.] 4 [ENTER] [.] 8 [.] 9 [×]

**B7: 10 gallons**
Setup: (24 × 5/8) × 2/3
SC-6: 24 [ENTER] [.] 5 [.] 8 [×] [.] 2 [.] 3 [×]

**B8: 1/8 in²**
Setup: 1/2²
SC-6: [.] 1 [.] 2 [x²]

**B9: 8 ft²**
Setup: 3 1/2 × 2 2/7
SC-6: 3 [.] 1 [.] 2 [ENTER] 2 [.] 2 [.] 7 [×]

**B10: 1/12 of the day (2 hours)**
Setup: 1/3 × 1/4
SC-6: [.] 1 [.] 3 [ENTER] [.] 1 [.] 4 [×]

## SEC 2.3 (pp.19-20): DIVIDING FRACTIONS AND MIXED NUMBERS [FRAC-01]

Methods: W1: Multiply by the Reciprocal | W2: Common-Denominator Division | W3: Fraction Strip / Number-Line Model

### PRACTICE A (p.19): KEY IDEAS (#1-#10) — ExA: 6

**A1: 2**
Setup: 1/2 ÷ 1/4
SC-6: [.] 1 [.] 2 [ENTER] [.] 1 [.] 4 [÷]

**A2: 4**
Setup: 2/3 ÷ 1/6
SC-6: [.] 2 [.] 3 [ENTER] [.] 1 [.] 6 [÷]

**A3: 2**
Setup: 5/6 × 1/(5/12)
SC-6: [.] 5 [.] 6 [ENTER] [.] 5 [.] 12 [f][1/x] [×]

**A4: 8**
Setup: 6 ÷ 3/4
SC-6: 6 [ENTER] [.] 3 [.] 4 [÷]

**A5: 1 7/8 (15/8)**
Setup: 3/4 ÷ 2/5
SC-6: [.] 3 [.] 4 [ENTER] [.] 2 [.] 5 [÷]

**A6: 7/8 cup**
Setup: 3 1/2 ÷ 4
SC-6: 3 [.] 1 [.] 2 [ENTER] 4 [÷]

**A7: 2 2/3 bike frames**
Setup: 2/3 ÷ 1/4
SC-6: [.] 2 [.] 3 [ENTER] [.] 1 [.] 4 [÷]

**A8: 3**
Setup: 5 1/4 ÷ 1 3/4
SC-6: 5 [.] 1 [.] 4 [ENTER] 1 [.] 3 [.] 4 [÷]

**A9: 5 full slats (5 7/37 -> 3 1/2 in left)**
Setup: 96 ÷ 18 1/2
SC-6: 96 [ENTER] 18 [.] 1 [.] 2 [÷]

**A10: 6 glasses**
Setup: 3 3/4 ÷ 5/8
SC-6: 3 [.] 3 [.] 4 [ENTER] [.] 5 [.] 8 [÷]

### PRACTICE B (p.20): REAL LIFE (#1-#10) — ExB: 6 shelves

**B1: 6 slices**
Setup: 3/4 ÷ 1/8
SC-6: [.] 3 [.] 4 [ENTER] [.] 1 [.] 8 [÷]

**B2: 12 pebbles**
Setup: 4 1/2 ÷ 3/8
SC-6: 4 [.] 1 [.] 2 [ENTER] [.] 3 [.] 8 [÷]

**B3: 6 runners**
Setup: 3 3/4 ÷ 5/8
SC-6: 3 [.] 3 [.] 4 [ENTER] [.] 5 [.] 8 [÷]

**B4: 7 segments**
Setup: 5 1/4 ÷ 3/4
SC-6: 5 [.] 1 [.] 4 [ENTER] [.] 3 [.] 4 [÷]

**B5: 40 layers**
Setup: 6 ÷ 3/20
SC-6: 6 [ENTER] [.] 3 [.] 20 [÷]

**B6: 6 sections**
Setup: 7 1/2 ÷ 1 1/4
SC-6: 7 [.] 1 [.] 2 [ENTER] 1 [.] 1 [.] 4 [÷]

**B7: 2 meters**
Setup: 4 1/2 ÷ 2 1/4
SC-6: 4 [.] 1 [.] 2 [ENTER] 2 [.] 1 [.] 4 [÷]

**B8: 8 cups**
Setup: 2 2/3 ÷ 1/3
SC-6: 2 [.] 2 [.] 3 [ENTER] [.] 1 [.] 3 [÷]

**B9: 5/12 (12/5 × 5/12 = 1)**
Setup: 1/(2 2/5)
SC-6: 2 [.] 2 [.] 5 [f][1/x]

**B10: 12 pieces**
Setup: (10 1/2 - 1 1/2) ÷ 3/4
SC-6: 10 [.] 1 [.] 2 [ENTER] 1 [.] 1 [.] 2 [-] [.] 3 [.] 4 [÷]

## SEC 2.4 (pp.21-22): ADDING, SUBTRACTING, AND MULTIPLYING DECIMALS [PLACE-01]

Methods: W1: Vertical Place-Value Algorithm | W2: Convert to Fractions/Hundredths | W3: Place-Value Split / Area Model

### PRACTICE A (p.21): KEY IDEAS (#1-#10) — ExA: 18.45

**A1: 345.67 mm**
Setup: (345 + 0.6) + 0.07
SC-6: 345 [ENTER] 0.6 [+] 0.07 [+]

**A2: 17.55**
Setup: 12.8 + 4.75
SC-6: 12.8 [ENTER] 4.75 [+]

**A3: 11.35**
Setup: 20 - 8.65
SC-6: 20 [ENTER] 8.65 [-]

**A4: 2.7**
Setup: 4.5 × 0.6
SC-6: 4.5 [ENTER] 0.6 [×]

**A5: 18.74**
Setup: (25.4 - 9.86) + 3.2
SC-6: 25.4 [ENTER] 9.86 [-] 3.2 [+]

**A6: 8.1 (Estimate: 4 × 2 = 8)**
Setup: 3.6 × 2.25
SC-6: 3.6 [ENTER] 2.25 [×]

**A7: 0.06 (3/50)**
Setup: 0.15 × 0.4
SC-6: 0.15 [ENTER] 0.4 [×] [FDISP]

**A8: 35.79 g (35.7864)**
Setup: 28.86 × 1.24
SC-6: 28.86 [ENTER] 1.24 [×]

**A9: $6.65**
Setup: 20 - (8.75 + 4.60)
SC-6: 20 [ENTER] 8.75 [ENTER] 4.60 [+] [-]

**A10: 161 meters**
Setup: ((30 × 1.20) + (40 × 1.35)) + (50 × 1.42)
SC-6: 30 [ENTER] 1.20 [×] 40 [ENTER] 1.35 [×] [+] 50 [ENTER] 1.42 [×…

### PRACTICE B (p.22): REAL LIFE (#1-#10) — ExB: $8.10

**B1: $41.24**
Setup: 18.75 + 22.49
SC-6: 18.75 [ENTER] 22.49 [+]

**B2: $6.35**
Setup: 20 - 13.65
SC-6: 20 [ENTER] 13.65 [-]

**B3: 0.55 seconds faster**
Setup: 13.42 - 12.87
SC-6: 13.42 [ENTER] 12.87 [-]

**B4: $7.80**
Setup: 2.40 × 3.25
SC-6: 2.40 [ENTER] 3.25 [×]

**B5: 1.19 cm²**
Setup: 1.4 × 0.85
SC-6: 1.4 [ENTER] 0.85 [×]

**B6: $41.40**
Setup: 11.5 × 3.60
SC-6: 11.5 [ENTER] 3.60 [×]

**B7: 30.35 meters**
Setup: (24.8 - 6.35) + 11.9
SC-6: 24.8 [ENTER] 6.35 [-] 11.9 [+]

**B8: $35.75**
Setup: (2 × 14.50) + 6.75
SC-6: 2 [ENTER] 14.50 [×] 6.75 [+]

**B9: 2.5 mm gap**
Setup: 170.5 - (4 × 42)
SC-6: 170.5 [ENTER] 4 [ENTER] 42 [×] [-]

**B10: $35.00 profit**
Setup: (3.20 - 1.80) × 25
SC-6: 3.20 [ENTER] 1.80 [-] 25 [×]

## SEC 2.5 (pp.23-24): DIVIDING WHOLE NUMBERS AND DECIMALS [FRAC-01]

Methods: W1: Power-of-10 Decimal Shift & Long Division | W2: Equivalent Fraction Scaling | W3: Reciprocal Fraction Division

### PRACTICE A (p.23): KEY IDEAS (#1-#10) — ExA: 5 R 3 = 5 3/4 = 5.75

**A1: 4 R 7, 4 7/8, and 4.875**
Setup: 39 ÷ 8
SC-6: 39 [ENTER] 8 [÷] [FDISP]

**A2: 2.3**
Setup: 18.4 ÷ 8
SC-6: 18.4 [ENTER] 8 [÷]

**A3: 45 quarters**
Setup: 11.25 ÷ 0.25
SC-6: 11.25 [ENTER] 0.25 [÷]

**A4: 8**
Setup: 7.2 ÷ 0.9
SC-6: 7.2 [ENTER] 0.9 [÷]

**A5: 130**
Setup: 15.6 ÷ 0.12
SC-6: 15.6 [ENTER] 0.12 [÷]

**A6: 6.3 (6 3/10)**
Setup: 9.45 ÷ 1.5
SC-6: 9.45 [ENTER] 1.5 [÷] [FDISP]

**A7: $20.80**
Setup: 124.80 ÷ 6
SC-6: 124.80 [ENTER] 6 [÷]

**A8: 24 (24 × 0.2 = 4.8)**
Setup: 4.8 ÷ 0.2
SC-6: 4.8 [ENTER] 0.2 [÷]

**A9: 6 laps**
Setup: 4.5 ÷ 0.75
SC-6: 4.5 [ENTER] 0.75 [÷]

**A10: $1.25 per bottle; 16 bottles**
Setup: 7.50 ÷ 6; 20 ÷ 1.25
SC-6: 7.50 [ENTER] 6 [÷] -> 1.25; 20 [ENTER] 1.25 [÷]

### PRACTICE B (p.24): REAL LIFE (#1-#10) — ExB: 27 quarters

**B1: 4 R 3 = 4 3/8 = 35/8 = 4.375**
Setup: 35 ÷ 8
SC-6: 35 [ENTER] 8 [÷] [FDISP]

**B2: 16 quarters**
Setup: 28 ÷ 1.75
SC-6: 28 [ENTER] 1.75 [÷]

**B3: $8.65 each**
Setup: 34.60 ÷ 4
SC-6: 34.60 [ENTER] 4 [÷]

**B4: 72 layers**
Setup: 14.4 ÷ 0.20
SC-6: 14.4 [ENTER] 0.20 [÷]

**B5: 15 flowers**
Setup: 5.25 ÷ 0.35
SC-6: 5.25 [ENTER] 0.35 [÷]

**B6: 6 miles per hour**
Setup: 4.5 ÷ 0.75
SC-6: 4.5 [ENTER] 0.75 [÷]

**B7: 18 sections**
Setup: 22.5 ÷ 1.25
SC-6: 22.5 [ENTER] 1.25 [÷]

**B8: 30 cups**
Setup: 13.5 ÷ 0.45
SC-6: 13.5 [ENTER] 0.45 [÷]

**B9: 7 tickets**
Setup: 87.50 ÷ 12.50
SC-6: 87.50 [ENTER] 12.50 [÷]

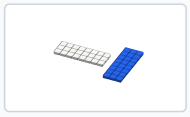
**B10: 6**
Setup: 10.5 ÷ 1.75
SC-6: 10.5 [ENTER] 1.75 [÷]

# Unit 3: Ratios, Rates & Percents — Practice A & B Complete Key

**Big Ideas Math Grade 6 — Chapters 3, 4 & 5 (Ratios, Rates & Percents) • Manipulative Kits: Counting & Array Tiles (COUNT-01) + Fraction Studio (FRAC-01)**

**Target Misconception: Students confuse Part-to-Part (3:2) with Part-to-Whole (3/5), add instead of multiply in ratio tables, divide unit rates upside-down, and memorize 6 disconnected rules for moving decimal points in percents.**

Hands-On Block Remediation: Use Blue and Gold Counting Tiles as physical Tape Diagram boxes, step through ratio tables rhythmically with [g] [REP], use [x↔y] when unit-rate division is entered upside-down, and use [f] [%] + [FDISP] to show Fraction/Decimal/Percent are three costumes of one number.



**QUEST-01 Double-Points Scoring: 1 '+10' token = Textbook Method 1 (1st Way); stack a 2nd '+10' token (+20 pts) = Textbook Method 2 (2nd Way)!**

## SEC 3.1 (pp.26-27): RATIOS AND TAPE DIAGRAMS [COUNT-01]

Methods: W1: Tape Diagram (Equal Boxes) | W2: Equivalent Ratio Fraction | W3: Divide by GCF

### PRACTICE A (p.26): KEY IDEAS (#1-#10)

ExA: Part-to-Part = 3:2; Part-to-Whole = 3:5

**A1: 2 : 3**
Setup: Ratio(12:18)
SC-6: 12 [ENTER] 18 [g][a:b]

**A2: 3 : 2**
Setup: Ratio(24:16)
SC-6: 24 [ENTER] 16 [g][a:b]

**A3: 2 : 5**
Setup: Ratio(4:10)
SC-6: 4 [ENTER] 10 [g][a:b]

**A4: 18 and 24 (Total = 42)**
Setup: 4 × 6
SC-6: 3 [ENTER] 6 [×], 4 [ENTER] 6 [×]

**A5: 4 L concentrate, 30 L water**
Setup: 34 ÷ 17
SC-6: 34 [ENTER] 17 [÷] -> 2/part

**A6: 20 dogs**
Setup: (36 ÷ 9) × 5
SC-6: 36 [ENTER] 9 [÷] 5 [×]

**A7: 25 seventh graders**
Setup: (15 ÷ 3) × 5
SC-6: 15 [ENTER] 3 [÷] 5 [×]

**A8: 30 electric guitars (12 acoustic)**
Setup: (18 ÷ 3) × 5
SC-6: 18 [ENTER] 3 [÷] 5 [×]

**A9: 48 gears and 36 gears**
Setup: 84 ÷ 7
SC-6: 84 [ENTER] 7 [÷] -> 12/box

**A10: 77 coins total (49 gold, 28 silver)**
Setup: (21 ÷ 3) × 11
SC-6: 21 [ENTER] 3 [÷] 11 [×]

### PRACTICE B (p.27): REAL LIFE (#1-#10)

ExB: 20 wins (and 12 losses)

**B1: Part-to-Part = 3:2; Part-to-Whole = 3:5**
Setup: Ratio(15:10)
SC-6: 15 [ENTER] 10 [g][a:b]

**B2: 16 strawberries, 12 bananas**
Setup: (28 ÷ 7) × 4
SC-6: 28 [ENTER] 7 [÷] 4 [×]

**B3: 6 cups yellow (21 cups total)**
Setup: (15 ÷ 5) × 2
SC-6: 15 [ENTER] 5 [÷] 2 [×]

**B4: 35 secret and 25 open**
Setup: (60 ÷ 12) × 7
SC-6: 60 [ENTER] 12 [÷] 7 [×]

**B5: 42 gold coins, 18 silver coins**
Setup: (24 ÷ 4) × 7
SC-6: 24 [ENTER] 4 [÷] 7 [×]

**B6: 3 : 1**
Setup: Ratio(48:16)
SC-6: 48 [ENTER] 16 [g][a:b]

**B7: 4 : 9 (16 out of 36)**
Setup: Ratio(16:36)
SC-6: 16 [ENTER] 36 [g][a:b]

**B8: 16 : 9**
Setup: Ratio(1920:1080)
SC-6: 1920 [ENTER] 1080 [g][a:b]

**B9: 150g almonds, 100g cashews, 250g raisins**
Setup: 500 ÷ 10
SC-6: 500 [ENTER] 10 [÷]

**B10: 3 : 4**
Setup: Ratio(24:32)
SC-6: 24 [ENTER] 32 [g][a:b]

## SEC 3.2 (pp.28-29): USING RATIO TABLES AND DOUBLE NUMBER LINES [COUNT-01]

Methods: W1: Multiplicative Scale Factor | W2: Double Number Line / Repeated Addition | W3: Scale Down Then Up

### PRACTICE A (p.28): KEY IDEAS (#1-#10)

ExA: 40 meters

**A1: 4:9, 8:18, 12:27, 16:36**
Setup: Repeat(+4), Repeat(+9)
SC-6: 4 [g][REP] [÷], 9 [g][REP] [÷]

**A2: 63**
Setup: (28 ÷ 4) × 9
SC-6: 28 [ENTER] 4 [÷] 9 [×]

**A3: 25**
Setup: (60 ÷ 12) × 5
SC-6: 60 [ENTER] 12 [÷] 5 [×]

**A4: 60 pages**
Setup: (8 ÷ 2) × 15
SC-6: 8 [ENTER] 2 [÷] 15 [×]

**A5: 48 ounces**
Setup: (18 ÷ 3) × 8
SC-6: 18 [ENTER] 3 [÷] 8 [×]

**A6: $15**
Setup: (12 ÷ 4) × 5
SC-6: 12 [ENTER] 4 [÷] 5 [×]

**A7: 84 beats**
Setup: (60 ÷ 15) × 21
SC-6: 60 [ENTER] 15 [÷] 21 [×]

**A8:** Ratios scale by multiplication (e.g., ×3 gives 9:15), not ad..
Setup: Ratio(7:9)
SC-6: 7 [ENTER] 9 [g][a:b]

**A9: 20 cups**
Setup: (12 ÷ 3) × 5
SC-6: 12 [ENTER] 3 [÷] 5 [×]

**A10: 40 miles**
Setup: (14 ÷ 7) × 20
SC-6: 14 [ENTER] 7 [÷] 20 [×]

### PRACTICE B (p.29): REAL LIFE (#1-#10)

ExB: $75

**B1: 45 grams**
Setup: 15 × 3
SC-6: 15 [ENTER] 3 [×]

**B2: 56 scoops**
Setup: (16 ÷ 2) × 7
SC-6: 16 [ENTER] 2 [÷] 7 [×]

**B3: $15 for 10 notebooks**
Setup: (12 ÷ 4) × 5
SC-6: 12 [ENTER] 4 [÷] 5 [×]

**B4: 112 beats per minute**
Setup: 28 × 4
SC-6: 28 [ENTER] 4 [×]

**B5: 260 gallons of sap**
Setup: 40 × 6.5
SC-6: 40 [ENTER] 6.5 [×]

**B6: 25 pounds**
Setup: 150 ÷ 6
SC-6: 150 [ENTER] 6 [÷]

**B7: 15 cups of flour**
Setup: (6 ÷ 2) × 5
SC-6: 6 [ENTER] 2 [÷] 5 [×]

**B8: 75 minutes (1 hr 15 min)**
Setup: (90 ÷ 18) × 15
SC-6: 90 [ENTER] 18 [÷] 15 [×]

**B9: 8 inches**
Setup: (60 ÷ 15) × 2
SC-6: 60 [ENTER] 15 [÷] 2 [×]

**B10: 30 lemons**
Setup: (9 ÷ 3) × 10
SC-6: 9 [ENTER] 3 [÷] 10 [×]

## SEC 3.3 (pp.30-31): RATES, UNIT RATES, AND COMPARING RATES [COUNT-01]

Methods: W1: Divide to Find the Unit Rate (1 Unit) | W2: Common Multiple Comparison | W3: Double Number Line

### PRACTICE A (p.30): KEY IDEAS (#1-#10)

ExA: 63 words per minute

**A1: 65 mph**
Setup: 195 ÷ 3
SC-6: 195 [ENTER] 3 [÷]

**A2: $9 per hour**
Setup: 36 ÷ 4
SC-6: 36 [ENTER] 4 [÷]

**A3: $1.30 per pound**
Setup: 7.80 ÷ 6
SC-6: 7.80 [ENTER] 6 [÷]

**A4: 15 grams per hour**
Setup: 90 ÷ 6
SC-6: 90 [ENTER] 6 [÷]

**A5: Supplier B ($0.17/oz vs. $0.18/oz)**
Setup: 3.24 ÷ 18; 4.08 ÷ 24
SC-6: 3.24 [ENTER] 18 [÷]; 4.08 [ENTER] 24 [÷]

**A6: 10-pack ($0.82/pen vs. $0.85/pen)**
Setup: 5.10 ÷ 6; 8.20 ÷ 10
SC-6: 5.10 [ENTER] 6 [÷]; 8.20 [ENTER] 10 [÷]

**A7: 300 miles**
Setup: (210 ÷ 3.5) × 5
SC-6: 210 [ENTER] 3.5 [÷] 5 [×]

**A8: Pump A (9 gal/min vs. 8 gal/min)**
Setup: 45 ÷ 5; 64 ÷ 8
SC-6: 45 [ENTER] 5 [÷]; 64 [ENTER] 8 [÷]

**A9: 180 mm**
Setup: (45 ÷ 15) × 60
SC-6: 45 [ENTER] 15 [÷] 60 [×]

**A10: Car B by 1 mpg (29 mpg vs. 28 mpg)**
Setup: 168 ÷ 6; 232 ÷ 8
SC-6: 168 [ENTER] 6 [÷]; 232 [ENTER] 8 [÷]

### PRACTICE B (p.31): REAL LIFE (#1-#10)

ExB: Pack B ($1.40 per battery vs. $1.45 per battery)

**B1: 150 miles per hour**
Setup: 450 ÷ 3
SC-6: 450 [ENTER] 3 [÷]

**B2: Box B ($0.70/bar vs. $0.75/bar)**
Setup: 7.00 ÷ 10
SC-6: 4.50 [ENTER] 6 [÷] 7.00 [ENTER] 10 [÷]

**B3: Spool B ($1.95 vs. $2.20 per 100g)**
Setup: 19.50 ÷ 10
SC-6: 11 [ENTER] 5 [÷] 19.50 [ENTER] 10 [÷]

**B4: 15 gal/min; 375 gallons**
Setup: (180 ÷ 12) × 25
SC-6: 180 [ENTER] 12 [÷] 25 [×]

**B5: B (57 taps/min vs. A's 55 taps/min)**
Setup: 285 ÷ 5
SC-6: 220 [ENTER] 4 [÷] 285 [ENTER] 5 [÷]

**B6: Car A (28 mpg vs. 27 mpg)**
Setup: 405 ÷ 15
SC-6: 336 [ENTER] 12 [÷] 405 [ENTER] 15 [÷]

**B7: $9 per dog -> 11 dogs**
Setup: 99 ÷ (54 ÷ 6)
SC-6: 54 [ENTER] 6 [÷] 99 [x⇔y] [÷]

**B8: 12-pack ($0.42 vs. $0.45 each)**
Setup: 5.04 ÷ 12
SC-6: 3.60 [ENTER] 8 [÷] 5.04 [ENTER] 12 [÷]

**B9: 4 inches per hour**
Setup: 14 ÷ 3.5
SC-6: 14 [ENTER] 3.5 [÷]

**B10: 4.5 m/s -> 90 meters**
Setup: (27 ÷ 6) × 20
SC-6: 27 [ENTER] 6 [÷] 20 [×]

## SEC 3.4 (pp.32-33): PERCENTS, FRACTIONS, AND DECIMALS [FRAC-01]

Methods: W1: Equivalent Fraction over 100 | W2: Divide Numerator ÷ Denominator | W3: Percent Bar / Benchmark Model

### PRACTICE A (p.32): KEY IDEAS (#1-#10)

ExA: 0.35 and 35%

**A1: 0.6 and 60%**
Setup: 3/5 -> Dec/Frac
SC-6: [.] 3 [.] 5 [ENTER] [FDISP]

**A2: 0.36 and 36%**
Setup: 9/25 -> Dec/Frac
SC-6: [.] 9 [.] 25 [ENTER] [FDISP]

**A3: 0.45 and 9/20**
Setup: 45 ÷ 100
SC-6: 45 [ENTER] 100 [÷] [FDISP]

**A4: 12.5% and 1/8**
Setup: 0.125 -> Dec/Frac
SC-6: 0.125 [ENTER] [FDISP]

**A5: 270 seats**
Setup: 60% of 450
SC-6: 450 [ENTER] 60 [%]

**A6: 3,510 grams**
Setup: 65% of 5400
SC-6: 5400 [ENTER] 65 [%]

**A7: 84%**
Setup: 42 ÷ 50
SC-6: 42 [ENTER] 50 [÷]

**A8: 1.5 and 1 1/2**
Setup: 150 ÷ 100
SC-6: 150 [ENTER] 100 [÷] [FDISP]

**A9: 60%, 0.61, 5/8**
Setup: 5/8 -> Dec/Frac
SC-6: [.] 5 [.] 8 [FDISP] -> 0.625

**A10: 1,400 mAh**
Setup: 35% of 4000
SC-6: 4000 [ENTER] 35 [%]

### PRACTICE B (p.33): REAL LIFE (#1-#10)

ExB: 75% (3/4 = 0.75)

**B1: 0.85 = 85%**
Setup: 17/20 × 100
SC-6: [.] 17 [.] 20 [FDISP] 100 [×]

**B2: 0.36 = 36%**
Setup: 9/25 × 100
SC-6: [.] 9 [.] 25 [FDISP] 100 [×]

**B3: 20 + 40 = 60 points**
Setup: 40% of 50 + 80% of 50
SC-6: 50 [ENTER] 40 [%] 50 [ENTER] 80 [%] [+]

**B4: 104 students**
Setup: 65% of 160
SC-6: 160 [ENTER] 65 [%]

**B5: 0.15 and 3/20**
Setup: Simp(15/100)
SC-6: [.] 15 [.] 100 [g][SIMP]

**B6: 45 students total**
Setup: 18 ÷ 0.40
SC-6: 18 [ENTER] 0.40 [÷]

**B7: $9.00**
Setup: 15% of 60
SC-6: 60 [ENTER] 15 [%]

**B8: 7/8 = 0.875 = 87.5%**
Setup: 21 ÷ 24
SC-6: 21 [ENTER] 24 [÷] [FDISP]

**B9: $480 goal**
Setup: 360 ÷ 0.75
SC-6: 360 [ENTER] 0.75 [÷]

**B10: 58% < 3/5 (60%) < 0.62 (62%)**
Setup: 3/5 -> Dec/Frac
SC-6: [.] 3 [.] 5 [FDISP]

## SEC 3.5 (pp.34-35): SOLVING PERCENT PROBLEMS, DISCOUNTS & CONVERSIONS [PLACE-01]

Methods: W1: Find Discount & Subtract | W2: Multiply by Remaining % | W3: Percent Proportion / Ratio Bar

### PRACTICE A (p.34): KEY IDEAS (#1-#10)

ExA: Discount = $15; Sale Price = $45

**A1: Discount $17; Sale price $68**
Setup: 20% of 85
SC-6: 85 [ENTER] 20 [%] [-]

**A2: Tax $10.20; Total $130.20**
Setup: 8.5% of 120
SC-6: 120 [ENTER] 8.5 [%] [+]

**A3: +400% ((75-15)÷15 = 400%)**
Setup: %Chg(15->75)
SC-6: 15 [ENTER] 75 [g][%CHG]

**A4: 25% decrease**
Setup: %Chg(80->60)
SC-6: 80 [ENTER] 60 [g][%CHG]

**A5: 25.4 cm**
Setup: 10 (to cm)
SC-6: 10 [f][->cm]

**A6: 20 in**
Setup: 50.8 (to in)
SC-6: 50.8 [g][->in]

**A7: $39.20**
Setup: 30% of 56
SC-6: 56 [ENTER] 30 [%] [-]

**A8: 72**
Setup: 18 ÷ 0.25
SC-6: 18 [ENTER] 0.25 [÷]

**A9: $144 ($200 -> $160 -> $144)**
Setup: 10% of 20% of 200
SC-6: 200 [ENTER] 20 [%] [-] 10 [%] [-]

**A10: 81.28 cm and 2.27 kg**
Setup: 32 (to cm); 5 (to kg)
SC-6: 32 [f][->cm]; 5 [f][->kg]

### PRACTICE B (p.35): REAL LIFE (#1-#10)

ExB: 30% increase; 81.28 cm

**B1: Discount = $15; Sale Price = $60**
Setup: 20% of 75
SC-6: 75 [ENTER] 20 [%] [-]

**B2: Tip = $9.60; Total = $73.60**
Setup: 15% of 64
SC-6: 64 [ENTER] 15 [%] [+]

**B3: Tax = $3.60; Total = $51.60**
Setup: 7.5% of 48
SC-6: 48 [ENTER] 7.5 [%] [+]

**B4: 30% increase**
Setup: %Chg(40->52)
SC-6: 40 [ENTER] 52 [g][%CHG]

**B5: 30% decrease**
Setup: %Chg(90->63)
SC-6: 90 [ENTER] 63 [g][%CHG]

**B6: 16.09 km**
Setup: 10 × 1.609344
SC-6: 10 [ENTER] 1.609344 [×]

**B7: 44.09 lb**
Setup: 20 (to lb)
SC-6: 20 [g][->lb]

**B8: 3.79 liters; 128 fl oz**
Setup: 1 (to L)
SC-6: 1 [f][->L]

**B9: $72 (10% is taken off $80, not $100)**
Setup: 10% of 20% of 100
SC-6: 100 [ENTER] 20 [%] [-] 10 [%] [-]

**B10: 68°F**
Setup: 20 (to °F)
SC-6: 20 [g][->°F]

# Unit 4: Algebraic Expressions & Properties — Practice A & B Complete Key

**Big Ideas Math Grade 6 — Chapter 3 / Chapter 5 (Algebraic Expressions) • Manipulative Kits: Algebra Boxes & Factoring Stage (ALGB-01) + Counting Tiles (COUNT-01)**

**Target Misconception: Students panic when letters appear, erroneously combine 3x + 2 into 5x or $x^2 + x$ into $x^3$, and don't see why 3(x + 4) = 3x + 12.**

Hands-On Block Remediation: Show that a variable x is literally a physical green x-rod (42×14mm) whereas 1 is a small yellow unit cube (14×14mm) and $x^2$ is a large blue flat (42×42mm)—different physical shapes cannot be counted as one pile! Build rectangles on the L-frame stage for the Distributive Property.



**QUEST-01 Double-Points Scoring: 1 '+10' token = Textbook Method 1 (1st Way); stack a 2nd '+10' token (+20 pts) = Textbook Method 2 (2nd Way)!**

## SEC 4.1 (pp.37-38): WRITING AND EVALUATING ALGEBRAIC EXPRESSIONS [ALGB-01]

Methods: W1: Substitute in Parentheses & Follow PEMDAS | W2: Equal-Groups Bar / Algebra Tile Model

### PRACTICE A (p.37): KEY IDEAS (#1-#10) — ExA: 27

**A1: 33**
Setup: (6 × 7) - 9
SC-6: 6 [ENTER] 7 [×] 9 [-]

**A2: 54**
Setup: (5 × 8) + 14
SC-6: 5 [ENTER] 8 [×] 14 [+]

**A3: 13**
Setup: 40 - (3 × 9)
SC-6: 40 [ENTER] 3 [ENTER] 9 [×] [-]

**A4: 47**
Setup: $6^2$ + 11
SC-6: 6 [x²] 11 [+]

**A5: 27**
Setup: ($4^2$ × 2) - 5
SC-6: 4 [x²] 2 [×] 5 [-]

**A6: 55**
Setup: (7 × 4) + (3 × 9)
SC-6: 7 [ENTER] 4 [×] 3 [ENTER] 9 [×] [+]

**A7: 8**
Setup: (19 + 13) ÷ 4
SC-6: 19 [ENTER] 13 [+] 4 [÷]

**A8: 15.1**
Setup: (5 × 3.5) - 2.4
SC-6: 5 [ENTER] 3.5 [×] 2.4 [-]

**A9: 8 + 4.50h; $26**
Setup: (4.50 × 4) + 8
SC-6: 4.50 [ENTER] 4 [×] 8 [+]

**A10: 150 $cm^2$**
Setup: $5^2$ × 6
SC-6: 5 [x²] 6 [×]

### PRACTICE B (p.38): REAL LIFE (#1-#10) — ExB: 99 points

**B1: 5g + 6; $26**
Setup: (5 × 4) + 6
SC-6: 5 [ENTER] 4 [×] 6 [+]

**B2: 50 - 4r; 14 tickets**
Setup: 50 - (4 × 9)
SC-6: 50 [ENTER] 4 [ENTER] 9 [×] [-]

**B3: $225**
Setup: ($5^2$ × 8) + 25
SC-6: 5 [x²] 8 [×] 25 [+]

**B4: 82 points**
Setup: (10 × 7) + (3 × 4)
SC-6: 10 [ENTER] 7 [×] 3 [ENTER] 4 [×] [+]

**B5: 144 feet**
Setup: $3^3$ × 16
SC-6: 3 [x³] 16 [×]

**B6: 96 $in^2$**
Setup: $4^2$ × 6
SC-6: 4 [x²] 6 [×]

**B7: 7p + 4d; $41**
Setup: (7 × 3) + (4 × 5)
SC-6: 7 [ENTER] 3 [×] 4 [ENTER] 5 [×] [+]

**B8: 78**
Setup: A = ½((11 + 15) × 6)
SC-6: 11 [ENTER] 15 [+] 6 [g][TRI]

**B9: 111 minutes (1 hr 51 min)**
Setup: (12 × 8) + 15
SC-6: 12 [ENTER] 8 [×] 15 [+]

**B10: 128 - 75 = 53**
Setup: ($4^3$ × 2) - ($5^2$ × 3)
SC-6: 4 [x³] 2 [×] 5 [x²] 3 [×] [-]

## SEC 4.2 (pp.39-40): SIMPLIFYING EXPRESSIONS BY COMBINING LIKE TERMS [ALGB-01]

Methods: W1: Reorder & Add Coefficients (Properties) | W2: Sort by Tile Shape / Column Alignment

### PRACTICE A (p.39): KEY IDEAS (#1-#10) — ExA: 6x + 8

**A1: 9x + 13**
Setup: 7 + 2; 4 + 9
SC-6: 7 [ENTER] 2 [+]; 4 [ENTER] 9 [+]

**A2: 5x + 7**
Setup: 8 - 3; 11 - 4
SC-6: 8 [ENTER] 3 [-]; 11 [ENTER] 4 [-]

**A3: 8y + 6**
Setup: 9 - 1
SC-6: 9 [ENTER] 1 [-]

**A4: 8a + 13**
Setup: 12 - 4; 5 + 8
SC-6: 12 [ENTER] 4 [-]; 5 [ENTER] 8 [+]

**A5: 6x² + 8x + 7**
Setup: 4+2=6; 3+5=8
SC-6: Mental / Table Check

**A6: 4n² + 5n + 10**
Setup: 6-2=4; 9-4=5
SC-6: Mental / Table Check

**A7: 5.3x + 2.7**
Setup: 3.5 + 1.8; 4.2 - 1.5
SC-6: 3.5 [ENTER] 1.8 [+]; 4.2 [ENTER] 1.5 [-]

**A8: 53 != 80; 5x + 3 is already in simplest form**
Setup: 5(10)+3=53 vs 8(10)=80
SC-6: Mental / Table Check

**A9: 6x + 16; 76 meters**
Setup: 2+2+1+1=6x; 3+3+5+5=16
SC-6: Mental / Table Check

**A10: 9x + 8; 53 cm**
Setup: 3+2+4=9x; 2+7-1=8
SC-6: Mental / Table Check

### PRACTICE B (p.40): REAL LIFE (#1-#10) — ExB: 5x² + 4x + 6

**B1: 9x + 9**
Setup: (3 + 1) + 5
SC-6: 2 [ENTER] 3 [+] 4 [+] 3 [ENTER] 1 [+] 5 [+]

**B2: 8x + 12**
Setup: (4 + 2) × 2
SC-6: 3 [ENTER] 1 [+] 2 [×] 4 [ENTER] 2 [+] 2 [×]

**B3: 4x + 14**
Setup: 9 + 5
SC-6: 6 [ENTER] 2 [-] 9 [ENTER] 5 [+]

**B4: 6x² + 4x + 11**
Setup: 3 + 8
SC-6: 4 [ENTER] 2 [+] 7 [ENTER] 3 [-] 3 [ENTER] 8 [+]

**B5: 13c + 31**
Setup: 12 + 19
SC-6: 8 [ENTER] 5 [+] 12 [ENTER] 19 [+]

**B6: 7x + 19 (coeff = 7, constant = 19)**
Setup: 14 + 5
SC-6: 9 [ENTER] 2 [-] 14 [ENTER] 5 [+]

**B7: (1/2)n + 13**
Setup: 4 + 9
SC-6: [.] 3 [.] 8 [ENTER] [.] 1 [.] 8 [+] 4 [ENTER] 9 [+]

**B8: 4y + 3.5**
Setup: 5.8 - 2.3
SC-6: 2.4 [ENTER] 1.6 [+] 5.8 [ENTER] 2.3 [-]

**B9: 6m + 9 (both equal 39 at m=5)**
Setup: ((10 - 4) × 5) + (15 - 6)
SC-6: 10 [ENTER] 4 [-] 5 [×] 15 [ENTER] 6 [-] [+]

**B10: 3x² + 7**
Setup: 10 - 3
SC-6: 5 [ENTER] 2 [-] 10 [ENTER] 3 [-]

## SEC 4.3 (pp.41-42): THE DISTRIBUTIVE PROPERTY [ALGB-01]

Methods: W1: Multiply Each Term Inside a(b + c) = ab + ac | W2: Split-Rectangle Area Model | W3: Repeated Addition of Rows

### PRACTICE A (p.41): KEY IDEAS (#1-#10) — ExA: 84

**A1: 140 + 21 = 161**
Setup: (7 × 20) + (7 × 3)
SC-6: 7 [ENTER] 20 [×] 7 [ENTER] 3 [×] [+]

**A2: 4x + 20**
Setup: 4 × 5
SC-6: 4 [ENTER] 5 [×]

**A3: 18x + 24**
Setup: 6×3=18; 6×4=24
SC-6: Mental / Table Check

**A4: 16y - 40**
Setup: 8×2=16; 8×5=40
SC-6: Mental / Table Check

**A5: 387**
Setup: (9 × 40) + (9 × 3)
SC-6: 9 [ENTER] 40 [×] 9 [ENTER] 3 [×] [+]

**A6: 17x + 6**
Setup: 3×4+5=17; 3×2=6
SC-6: Mental / Table Check

**A7: 12n + 13**
Setup: 4×3=12; 20-7=13
SC-6: Mental / Table Check

**A8: 5x + 30 (She forgot to multiply 6 by 5)**
Setup: 5 × 6
SC-6: 5 [ENTER] 6 [×]

**A9: 14x + 15 (Both equal 43 when x = 2)**
Setup: 5×2+4=14; 5×3=15
SC-6: Mental / Table Check

**A10: 6(x + 3.50) = 6x + 21; $60 total**
Setup: 6 × 3.50
SC-6: 6 [ENTER] 3.50 [×] -> 6x+21

### PRACTICE B (p.42): REAL LIFE (#1-#10) — ExB: 14x + 15

**B1: 8(50 + 3) = 400 + 24 = $424**
Setup: (8 × 50) + (8 × 3)
SC-6: 8 [ENTER] 50 [×] 8 [ENTER] 3 [×] [+]

**B2: 6(10 - 0.10) = $60.00 - $0.60 = $59.40**
Setup: (6 × 10) - (6 × 0.10)
SC-6: 6 [ENTER] 10 [×] 6 [ENTER] 0.10 [×] [-]

**B3: 7(x + 8) = 7x + 56**
Setup: 7 × 8
SC-6: 7 [ENTER] 8 [×]

**B4: 9(2x + 7) = 18x + 63**
Setup: 9 × 7
SC-6: 9 [ENTER] 2 [×] 9 [ENTER] 7 [×]

**B5: 4(t - 3) = 4t - 12 ($48 at t = $15)**
Setup: (4 × 15) - (4 × 3)
SC-6: 4 [ENTER] 15 [×] 4 [ENTER] 3 [×] [-]

**B6: 14x + 15**
Setup: 3 × 5
SC-6: 3 [ENTER] 4 [×] 2 [+] 3 [ENTER] 5 [×]

**B7: 13x + 22**
Setup: (4 × 3) + (5 × 2)
SC-6: 4 [ENTER] 2 [×] 5 [+] 4 [ENTER] 3 [×] 5 [ENTER] 2 [×] [+]

**B8: 3x + 5**
Setup: 10 ÷ 2
SC-6: 6 [ENTER] 2 [÷] 10 [ENTER] 2 [÷]

**B9: 2(2x + 5) + 2(x + 3) = 6x + 16**
Setup: (2 × 5) + (2 × 3)
SC-6: 2 [ENTER] 2 [×] 2 [+] 2 [ENTER] 5 [×] 2 [ENTER] 3 [×] [+]

**B10: 12(15) = 180 and 120 + 60 = 180**
Setup: 12 × 15
SC-6: 12 [ENTER] 15 [×]

## SEC 4.4 (pp.43-44): FACTORING EXPRESSIONS USING THE GCF [ALGB-01]

Methods: W1: Factor Out the GCF Using Prime Factors | W2: Equal-Rows Area Model / Division Ladder

### PRACTICE A (p.43): KEY IDEAS (#1-#10) — ExA: 6(3x + 4)

**A1: 4(3 + 5)**
Setup: GCF(12, 20)
SC-6: 12 [ENTER] 20 [GCF] -> 4

**A2: 4(2x + 3)**
Setup: GCF(8, 12)
SC-6: 8 [ENTER] 12 [GCF] -> 4

**A3: 5(3y - 5)**
Setup: GCF(15, 25)
SC-6: 15 [ENTER] 25 [GCF] -> 5

**A4: 12(2x + 3)**
Setup: GCF(24, 36)
SC-6: 24 [ENTER] 36 [GCF] -> 12

**A5: 16(2n - 3)**
Setup: GCF(32, 48)
SC-6: 32 [ENTER] 48 [GCF] -> 16

**A6: 14(3a + 1)**
Setup: GCF(42, 14)
SC-6: 42 [ENTER] 14 [GCF] -> 14

**A7: 12(3m - 5)**
Setup: GCF(36, 60)
SC-6: 36 [ENTER] 60 [GCF] -> 12

**A8: 12(2x + 3) (4 and 6 still share a factor of 2)**
Setup: GCF(4, 6)
SC-6: 4 [ENTER] 6 [GCF] -> 2 left!

**A9: Height = 14 in.; Width = (2x + 3) in.**
Setup: GCF(28, 42)
SC-6: 28 [ENTER] 42 [GCF] -> 14

**A10: (9x + 12) feet**
Setup: 36 ÷ 4; 48 ÷ 4
SC-6: 36 [ENTER] 4 [÷]; 48 [ENTER] 4 [÷]

### PRACTICE B (p.44): REAL LIFE (#1-#10) — ExB: 15(3x - 2)

**B1: 8(2x + 3) -> 8 bags**
Setup: GCF(16, 24)
SC-6: 16 [ENTER] 24 [GCF]

**B2: 7(3x + 5) -> length = 3x + 5 m**
Setup: GCF(21, 35)
SC-6: 21 [ENTER] 35 [GCF]

**B3: 4(5x + 9) -> side = 5x + 9 in**
Setup: GCF(20, 36)
SC-6: 20 [ENTER] 36 [GCF]

**B4: 3(6y - 9) -> side = 6y - 9 cm**
Setup: GCF(18, 27)
SC-6: 18 [ENTER] 27 [GCF]

**B5: 8(5x + 8)**
Setup: GCF(40, 64)
SC-6: 40 [ENTER] 64 [GCF]

**B6: 18(2s + 3) -> 18 player kits**
Setup: GCF(36, 54)
SC-6: 36 [ENTER] 54 [GCF]

**B7: 6(2x + 3y + 5)**
Setup: GCF(GCF(12, 18), 30)
SC-6: 12 [ENTER] 18 [GCF] 30 [GCF]

**B8: 7(5d - 2) -> 5d - 2 per row**
Setup: GCF(35, 14)
SC-6: 35 [ENTER] 14 [GCF]

**B9: 24x + 36 = 12(2x + 3)**
Setup: GCF((10 + 14), (18 + 18))
SC-6: 10 [ENTER] 14 [+] 18 [ENTER] 18 [+] [GCF]

**B10: 9(3x - 2); at x=2 both equal 36**
Setup: GCF(27, 18)
SC-6: 27 [ENTER] 18 [GCF]

# Unit 5: Areas of Polygons & Elapsed Time — Practice A & B Complete Key

**Big Ideas Math Grade 6 — Chapter 4 (Areas of Polygons) • Manipulative Kits: Counting & Array Tiles (COUNT-01) + Planetary Geared Clock (TIME-01)**

**Target Misconception: Students multiply the slanted side of a parallelogram instead of the vertical height, forget to divide by 2 for triangles, lose track of multi-piece composite areas, and subtract clock times in base-10 instead of base-60.**

Hands-On Block Remediation: Slide the corner triangle of a parallelogram to form a straight [f] [RECT] rectangle, use the dedicated [g] [TRI] key ([g][7]) which divides by 2 automatically, accumulate composite chunks with [Σ+], and use the Planetary Clock (TIME-01) + [CLK] key for base-60 time.

**QUEST-01 Double-Points Scoring: 1 '+10' token = Textbook Method 1 (1st Way); stack a 2nd '+10' token (+20 pts) = Textbook Method 2 (2nd Way)!**



## SEC 5.1 (pp.46-47): AREAS OF PARALLELOGRAMS AND RECTANGLES [COUNT-01]

Methods: W1: Area Formula A = bh | W2: Cut-and-Slide Rectangle Decomposition

### PRACTICE A (p.46): KEY IDEAS (#1-#10) — ExA: 96 cm²

**A1: 135 m²**
Setup: A = 15 × 9
SC-6: 15 [ENTER] 9 [f][RECT]

**A2: 36 m²**
Setup: A = 8 × 4.5
SC-6: 8 [ENTER] 4.5 [f][RECT]

**A3: 54.4 m²**
Setup: A = 8.5 × 6.4
SC-6: 8.5 [ENTER] 6.4 [f][RECT]

**A4: 169 cm²**
Setup: 13²
SC-6: 13 [x²]

**A5: 26 in.²**
Setup: A = 6 1/2 × 4
SC-6: 6 [.] 1 [.] 2 [ENTER] 4 [f][RECT]

**A6: 9 cm**
Setup: 108 ÷ 12
SC-6: 108 [ENTER] 12 [÷]

**A7: 18 ft**
Setup: 135 ÷ 7.5
SC-6: 135 [ENTER] 7.5 [÷]

**A8: 60 cm² (He multiplied by the slanted side 7.5)**
Setup: A = 10 × 6
SC-6: 10 [ENTER] 6 [f][RECT]

**A9: 666 ft²**
Setup: (9×18.5) × 4
SC-6: 9 [ENTER] 18.5 [f][RECT] 4 [×]

**A10: 168 ft²**
Setup: (15×12) - (4×3)
SC-6: 15 [ENTER] 12 [f][RECT] 4 [ENTER] 3 [f][RECT] [-]

### PRACTICE B (p.47): REAL LIFE (#1-#10) — ExB: 12 inches

**B1: 180 ft²**
Setup: A = 10 × 18
SC-6: 10 [ENTER] 18 [f][RECT]

**B2: 60 ft**
Setup: (10 + 20) × 2
SC-6: 10 [ENTER] 20 [+] 2 [×]

**B3: 14.4 ft²**
Setup: A = 4.5 × 3.2
SC-6: 4.5 [ENTER] 3.2 [f][RECT]

**B4: 15 inches**
Setup: 135 ÷ 9
SC-6: 135 [ENTER] 9 [÷]

**B5: 216 cm²**
Setup: (6×4.5) × 8
SC-6: 6 [ENTER] 4.5 [f][RECT] 8 [×]

**B6: 228 ft²**
Setup: A = 19 × 12
SC-6: 19 [ENTER] 12 [f][RECT]

**B7: 2,520 ft²**
Setup: (65×40) - (10×8)
SC-6: 65 [ENTER] 40 [f][RECT] 10 [ENTER] 8 [f][RECT] [-]

**B8: 22 in²**
Setup: 5 1/2 × 4
SC-6: 5 [.] 1 [.] 2 [ENTER] 4 [×]

**B9: Both equal 144 cm² (same base and perpendicular height)**
Setup: A = 16 × 9
SC-6: 16 [ENTER] 9 [f][RECT]

**B10: 28 ft² × $4 = $112.00**
Setup: (8×3.5) × 4
SC-6: 8 [ENTER] 3.5 [f][RECT] 4 [×]

## SEC 5.2 (pp.48-49): AREAS OF TRIANGLES [COUNT-01]

Methods: W1: Formula A = (b × h) ÷ 2 | W2: Take Half of Base or Height First (½b × h)

### PRACTICE A (p.48): KEY IDEAS (#1-#10) — ExA: 30 cm²

**A1: 750 cm²**
Setup: A = ½(25 × 60)
SC-6: 25 [ENTER] 60 [g][TRI]

**A2: 18 ft²**
Setup: A = ½(9 × 4)
SC-6: 9 [ENTER] 4 [g][TRI]

**A3: 88 cm²**
Setup: A = ½(16 × 11)
SC-6: 16 [ENTER] 11 [g][TRI]

**A4: 52.5 m²**
Setup: A = ½(12.5 × 8.4)
SC-6: 12.5 [ENTER] 8.4 [g][TRI]

**A5: 13 1/2 in.² (13.5)**
Setup: A = ½(4 1/2 × 6)
SC-6: 4 [.] 1 [.] 2 [ENTER] 6 [g][TRI]

**A6: 9 cm**
Setup: (54 × 2) ÷ 12
SC-6: 54 [ENTER] 2 [×] 12 [÷]

**A7: 12 m**
Setup: (84 × 2) ÷ 14
SC-6: 84 [ENTER] 2 [×] 14 [÷]

**A8: 20 cm² (He forgot to divide by 2)**
Setup: A = ½(8 × 5)
SC-6: 8 [ENTER] 5 [g][TRI]

**A9: 210 ft²**
Setup: ½(12×25) + ½(8×15)
SC-6: 12 [ENTER] 25 [g][TRI] 8 [ENTER] 15 [g][TRI] [+]

**A10: 1,400 cm²**
Setup: A = ½(40 × 70)
SC-6: 40 [ENTER] 70 [g][TRI]

### PRACTICE B (p.49): REAL LIFE (#1-#10) — ExB: 8 inches

**B1: 150 ft²**
Setup: A = ½(12 × 25)
SC-6: 12 [ENTER] 25 [g][TRI]

**B2: 14 ft² (1 side); 28 ft² (both)**
Setup: A = ½(8 × 3.5)
SC-6: 8 [ENTER] 3.5 [g][TRI]

**B3: 141 ft²**
Setup: A = ½(30 × 9.4)
SC-6: 30 [ENTER] 9.4 [g][TRI]

**B4: 36 inches**
Setup: (450 × 2) ÷ 25
SC-6: 450 [ENTER] 2 [×] 25 [÷]

**B5: 6 feet**
Setup: (18 × 2) ÷ 6
SC-6: 18 [ENTER] 2 [×] 6 [÷]

**B6: 10.125 in² (10 1/8 in²)**
Setup: A = ½(4.5 × 4.5)
SC-6: 4.5 [ENTER] 4.5 [g][TRI]

**B7: 54 ft²**
Setup: A = ½(9 × 12)
SC-6: 9 [ENTER] 12 [g][TRI]

**B8: 630 cm² (315 cm² per wing)**
Setup: ½(18×35) × 2
SC-6: 18 [ENTER] 35 [g][TRI] 2 [×]

**B9: 13 in²**
Setup: (6 1/2 × 4) ÷ 2
SC-6: 6 [.] 1 [.] 2 [ENTER] 4 [×] 2 [÷]

**B10: 64 ft²**
Setup: ½(5×6.4) × 4
SC-6: 5 [ENTER] 6.4 [g][TRI] 4 [×]

## SEC 5.3 (pp.50-51): AREAS OF TRAPEZOIDS AND COMPOSITE FIGURES [COUNT-01]

Methods: W1: Decompose & Add Sub-Areas | W2: Trapezoid Formula A = ½h(b1 + b2) | W3: Enclosing Box Minus Cutouts

### PRACTICE A (p.50): KEY IDEAS (#1-#10) — ExA: 126 ft²

**A1: 72 cm²**
Setup: A = ½((10 + 14) × 6)
SC-6: 10 [ENTER] 14 [+] 6 [g][TRI]

**A2: 135 ft²**
Setup: A = ½((18 + 12) × 9)
SC-6: 18 [ENTER] 12 [+] 9 [g][TRI]

**A3: 92 ft²**
Setup: Stats(10, 6, 8, 4)
SC-6: 10 [ENTER] 6 [f][RECT] [Σ+] 8 [ENTER] 4 [f][RECT] [Σ+] [f][Σx]

**A4: 120 in.²**
Setup: Stats(15, 6, 10, 6)
SC-6: 15 [ENTER] 6 [f][RECT] [Σ+] 10 [ENTER] 6 [g][TRI] [Σ+] [f][Σx]

**A5: 29.4 m²**
Setup: A = ½((8.5 + 5.5) × 4.2)
SC-6: 8.5 [ENTER] 5.5 [+] 4.2 [g][TRI]

**A6: 56 in.²**
Setup: Stats(10, 8, 6, 4)
SC-6: 10 [ENTER] 8 [f][RECT] [Σ+] 6 [ENTER] 4 [f][RECT] [f][Σ-] [f][Σx]

**A7: 10 cm**
Setup: (90 × 2) ÷ (11 + 7)
SC-6: 90 [ENTER] 2 [×] 11 [ENTER] 7 [+] [÷]

**A8: 88 ft²**
Setup: Stats(60, 5, 9, 4)
SC-6: 60 [Σ+] 5 [ENTER] 9 [+] 4 [g][TRI] [Σ+] [f][Σx]

**A9: 104 cm² (80 + 24 = 104)**
Setup: A = ½((10 + 16) × 8)
SC-6: 10 [ENTER] 16 [+] 8 [g][TRI]

**A10: 251 ft²**
Setup: Stats(300, 24, 25)
SC-6: 300 [Σ+] 24 [f][Σ-] 25 [f][Σ-] [f][Σx]

### PRACTICE B (p.51): REAL LIFE (#1-#10) — ExB: 1,008 square inches

**B1: 440 in²**
Setup: A = ½((16 + 28) × 20)
SC-6: 16 [ENTER] 28 [+] 20 [g][TRI]

**B2: 40 in² per side; 160 in² total**
Setup: A = ½((4 + 6) × 8)
SC-6: 4 [ENTER] 6 [+] 8 [g][TRI]

**B3: 84 + 50 = 134 ft²**
Setup: (14×6) + (10×5)
SC-6: 14 [ENTER] 6 [f][RECT] 10 [ENTER] 5 [f][RECT] [+]

**B4: 42 cm² (1 fin); 126 cm² (3 fins)**
Setup: A = ½((5 + 9) × 6)
SC-6: 5 [ENTER] 9 [+] 6 [g][TRI]

**B5: 90 + 48 = 138 in²**
Setup: (15×6) + ½(12×8)
SC-6: 15 [ENTER] 6 [f][RECT] 12 [ENTER] 8 [g][TRI] [+]

**B6: 1,152 - 36 = 1,116 in²**
Setup: (48×24) - 6²
SC-6: 48 [ENTER] 24 [f][RECT] 6 [x²] [-]

**B7: h = 8 ft**
Setup: 104 ÷ 13
SC-6: 104 [ENTER] 13 [÷]

**B8: 280 + 60 = 340 ft²**
Setup: (20×14) + ½(20×6)
SC-6: 20 [ENTER] 14 [f][RECT] 20 [ENTER] 6 [g][TRI] [+]

**B9: 416 - 200 = 216 m²**
Setup: (26×16) - (20×10)
SC-6: 26 [ENTER] 16 [f][RECT] 20 [ENTER] 10 [f][RECT] [-]

**B10: 42 cm²**
Setup: ½((4 + 8)×3.5) × 2
SC-6: 4 [ENTER] 8 [+] 3.5 [g][TRI] 2 [×]

## SEC 5.4 (pp.52-53): POLYGONS IN THE COORDINATE PLANE & ELAPSED TIME [TIME-01]

Methods: W1: Subtract Coordinates / Regroup 60 min = 1 hr | W2: Count Grid Squares / Time-Line Jumps

### PRACTICE A (p.52): KEY IDEAS (#1-#10) — ExA: Width = 7, Height = 5, Perimeter = 24, Area = 35

**A1: 8 units**
Setup: 11 - 3
SC-6: 11 [ENTER] 3 [-]

**A2: 54 square units**
Setup: A = (10 - 1) × (8 - 2)
SC-6: 10 [ENTER] 1 [-] 8 [ENTER] 2 [-] [f][RECT]

**A3: 10:58 AM**
Setup: 7.10 + 3.48
SC-6: [CLK] 7.10 [ENTER] 3.48 [+]

**A4: 2.5 hours; 4.75 hours**
Setup: 2.30 (to HR); 4.45 (to HR)
SC-6: 2.30 [f][->HR]; 4.45 [f][->HR]

**A5: 6 hours 50 minutes**
Setup: 15.05 - 8.15
SC-6: [CLK] 15.05 [ENTER] 8.15 [-]

**A6: 1 hour 45 minutes**
Setup: 1.75 (to HMS)
SC-6: 1.75 [g][->HMS]

**A7: 2 hours 35 minutes**
Setup: 11.25 - 8.50
SC-6: [CLK] 11.25 [ENTER] 8.50 [-]

**A8: 24 square units**
Setup: A = ½(8 × 6)
SC-6: 8 [ENTER] 6 [g][TRI]

**A9: 5.5 hours (5 hr 30 min); $101.75**
Setup: 5.5 × 18.50
SC-6: 5.5 [ENTER] 18.50 [×]

**A10: Perimeter = 420 yd; Area = 10,800 yd²**
Setup: A = 12 × 9
SC-6: 12 [ENTER] 9 [f][RECT]

### PRACTICE B (p.53): REAL LIFE (#1-#10) — ExB: 12:30 PM

**B1: W=10, H=6, P=32, Area=60**
Setup: A = (11 - 1) × (8 - 2)
SC-6: 11 [ENTER] 1 [-] 8 [ENTER] 2 [-] [f][RECT]

**B2: Base=8, Height=6, Area=24**
Setup: A = ½((10 - 2) × (7 - 1))
SC-6: 10 [ENTER] 2 [-] 7 [ENTER] 1 [-] [g][TRI]

**B3: 12:15 PM**
Setup: 8.35 + 3.40
SC-6: [CLK] 8.35 [ENTER] 3.40 [+]

**B4: 3 hours 35 minutes**
Setup: 14.20 - 10.45
SC-6: [CLK] 14.20 [ENTER] 10.45 [-]

**B5: 3 hours 50 minutes**
Setup: 1.52 + 1.58
SC-6: [CLK] 1.52 [ENTER] 1.58 [+]

**B6: 3.5 hr × $14 = $49.00**
Setup: 3.5 × 14
SC-6: 3.5 [ENTER] 14 [×]

**B7: 2 hours 45 minutes**
Setup: 2.75 (to HMS)
SC-6: 2.75 [g][->HMS]

**B8: 4 hours 30 minutes**
Setup: (1.25 + 2.15) + 0.50
SC-6: [CLK] 1.25 [ENTER] 2.15 [+] 0.50 [+]

**B9: (3, 10); Area = 36**
Setup: A = (9 - 3) × (10 - 4)
SC-6: 9 [ENTER] 3 [-] 10 [ENTER] 4 [-] [f][RECT]

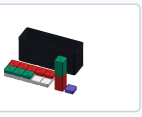
**B10: 180 miles**
Setup: 2.25 × 80
SC-6: 2.25 [ENTER] 80 [×]

# Unit 6: Integers, Rational Numbers & The Coordinate Plane — Practice A & B Complete Key

**Big Ideas Math Grade 6 — Chapter 6 • Manipulative Kits: Ten-Frame Two-Tone Counters (BASE-10) + Algebra Zero-Pair Tiles (ALGB-01)**

**Target Misconception: Students confuse the subtraction operator [-] with the negative sign [+/-], think -18 > -5 because 18 > 5, confuse Opposite ([+/-]) with Absolute Value ([g] [|x|]), and reverse (x, y) coordinates.**

Hands-On Block Remediation: Use two-tone (+1 / -1) Zero-Pair counters (BASE-10) and Zero-Pair tiles (ALGB-01), plot signed jumps on the SC-6 Number Line ([f] [LINE]), test inequalities with [f] [>] and [g] [<], contrast [+/-] (flip sign) with [g] [|x|] (always positive distance from 0), and use the Elevator Rule (walk across x first, ride up/down y second).



**QUEST-01 Double-Points Scoring: 1 '+10' token = Textbook Method 1 (1st Way); stack a 2nd '+10' token (+20 pts) = Textbook Method 2 (2nd Way)!**

## SEC 6.1 (pp.55-56): POSITIVE AND NEGATIVE INTEGERS & OPPOSITES [BASE-10]

Methods: W1: Equal Distance on Opposite Sides of 0 (Number Line) | W2: Additive Inverse & Zero-Pair Counters

### PRACTICE A (p.55): KEY IDEAS (#1-#10)

ExA: Opposite of -15 is +15; -15 + 15 = 0

**A1:** -25; Opposite = +25
Setup: -25; +/-
SC-6: 25 [+/-] -> -25; [+/-] -> 25

**A2:** +42; and +42
Setup: 42
SC-6: 42 [+/-] [+/-]

**A3:** 0 m (Sea level)
Setup: -18 + 18
SC-6: 18 [+/-] [ENTER] 18 [+]

**A4:** 0
Setup: 6 + -6
SC-6: 6 [ENTER] 6 [+/-] [+]

**A5:** 0 ft
Setup: -120 + 120
SC-6: 120 [+/-] [ENTER] 120 [+]

**A6:** 0 (0 is 0 units from itself)
Setup: -0
SC-6: 0 [+/-]

**A7:** +7°F
Setup: -12 + 19
SC-6: 12 [+/-] [ENTER] 19 [+]

**A8:** 0 (Even par)
Setup: -3 + 3
SC-6: 3 [+/-] [ENTER] 3 [+]

**A9:** Floor 3
Setup: (14 + -20) + 9
SC-6: 14 [ENTER] 20 [+/-] [+] 9 [+]

**A10:** +$18,800
Setup: ((14500 - 3200) + 8900) - 1400
SC-6: 14500 [ENTER] 3200 [-] 8900 [+] 1400 [-]

### PRACTICE B (p.56): REAL LIFE (#1-#10)

ExB: +$5

**B1:** 0 ft (opposites / additive inverses)
Setup: -120 + 120
SC-6: 120 [+/-] [ENTER] 120 [+]

**B2:** -282 ft; opposite is +282 ft
Setup: -282
SC-6: 282 [+/-]

**B3:** +19 yards
Setup: (12 + -8) + 15
SC-6: 12 [ENTER] 8 [+/-] [+] 15 [+]

**B4:** -28°C; opposite is +28°C
Setup: -28
SC-6: 28 [+/-]

**B5:** +$7
Setup: -18 + 25
SC-6: 18 [+/-] [ENTER] 25 [+]

**B6:** -5 (5 under par)
Setup: (-4 + 2) + -3
SC-6: 4 [+/-] [ENTER] 2 [+] 3 [+/-] [+]

**B7:** -x = +35; -(-x) = -35
Setup: 35
SC-6: 35 [+/-] [+/-]

**B8:** +$10
Setup: (25 - 25) + 10
SC-6: 25 [ENTER] 25 [-] 10 [+]

**B9:** +35 meters
Setup: (-40 + 95) - 20
SC-6: 40 [+/-] [ENTER] 95 [+] 20 [-]

**B10:** +23
Setup: -14 + -9
SC-6: 14 [+/-] [ENTER] 9 [+/-] [+] [+/-]

## SEC 6.2 (pp.57-58): COMPARING AND ORDERING INTEGERS & RATIONAL NUMBERS [FRAC-01]

Methods: W1: Convert to Decimals & Plot on Number Line | W2: Rewrite with Least Common Denominator (LCD)

### PRACTICE A (p.57): KEY IDEAS (#1-#10)

ExA: -5 > -18

**A1:** -5°C (-12 < -5)
Setup: Compare -5 < -12
SC-6: 5 [+/-] [ENTER] 12 [+/-] [g][<]

**A2:** -9 < -2
Setup: Compare -2 < -9
SC-6: 2 [+/-] [ENTER] 9 [+/-] [g][<]

**A3:** -15 < 0
Setup: Compare 0 < -15
SC-6: 0 [ENTER] 15 [+/-] [g][<]

**A4:** -10, -7, -2, 0, 4
Setup: fLINE
SC-6: Left-to-right on [f][LINE]

**A5:** -3/4 < -0.6 (-0.6 is greater)
Setup: Compare -0.6 < -0.75
SC-6: 0.6 [+/-] [ENTER] 0.75 [+/-] [g][<]

**A6:** -2 1/2 < -2.3
Setup: Compare -2.3 < -2.5
SC-6: 2.3 [+/-] [ENTER] 2.5 [+/-] [g][<]

**A7:** -5/8 > -7/8
Setup: Compare -7/8 > -5/8
SC-6: [.] 7 [.] 8 [+/-] [ENTER] [.] 5 [.] 8 [+/-] [f][>]

**A8:** -4.1 is closer to 0 (further to the right) than -4.5
Setup: Compare -4.5 < -4.1
SC-6: 4.5 [+/-] [ENTER] 4.1 [+/-] [g][<]

**A9:** -1 3/4, -1.5, -0.5, 1 1/4
Setup: -1.75, -1.5, -0.5, 1.25
SC-6: Mental / Table Check

**A10:** Deepest: -15.2 m; Shallowest: -14 1/2 m
Setup: -15.2 < -14.8 < -14.5
SC-6: Mental / Table Check

### PRACTICE B (p.58): REAL LIFE (#1-#10)

ExB: -0.75 > -4/5 (Diver B is closer to the surface)

**B1:** -11°C > -18°C (Tuesday was warmer)
Setup: Compare -18 > -11
SC-6: 18 [+/-] [ENTER] 11 [+/-] [f][>]

**B2:** Puffin (-45.8 m) is deeper; -45.25 > -45.8
Setup: Compare -45.8 > -45.25
SC-6: 45.8 [+/-] [ENTER] 45.25 [+/-] [f][>]

**B3:** -2/3 > -5/6 (-4/6 > -5/6)
Setup: Compare -5/6 > -2/3
SC-6: [.] 5 [.] 6 [+/-] [ENTER] [.] 2 [.] 3 [+/-] [f][>]

**B4:** -9, -6, -1, +2
Setup: Compare -6 < -9
SC-6: 6 [+/-] [ENTER] 9 [+/-] [g][<]

**B5:** -89°C, -39°C, -6°C, 0°C
Setup: Compare -39 < -89
SC-6: 39 [+/-] [ENTER] 89 [+/-] [g][<]

**B6:** Sea turtle (-3.1 > -3.25)
Setup: Compare -3.1 < -3.25
SC-6: 3.1 [+/-] [ENTER] 3.25 [+/-] [g][<]

**B7:** Stock B (-1.08% > -1.8%)
Setup: Compare -1.8 > -1.08
SC-6: 1.8 [+/-] [ENTER] 1.08 [+/-] [f][>]

**B8:** Examples: -3 1/2 and -3.5
Setup: -3.5
SC-6: 3.5 [+/-]

**B9:** -2, -7/4 (-1.75), -1.6, 3/2 (1.5)
Setup: -7/4 -> Dec/Frac
SC-6: [.] 7 [.] 4 [+/-] [FDISP]

**B10:** a is farther from 0 (e.g., -10 < -2)
Setup: |-10|
SC-6: 10 [+/-] [g][|x|]

## SEC 6.3 (pp.59-60): ABSOLUTE VALUE AND MAGNITUDE [ALGB-01]

Methods: W1: Measure Distance from 0 on a Number Line | W2: Evaluate |x| Grouping Bars Before Outside Signs

### PRACTICE A (p.59): KEY IDEAS (#1-#10)

ExA: |-14| = 14, |14| = 14, -|-14| = -14

**A1:** 125
Setup: |-125|
SC-6: 125 [+/-] [g][|x|]

**A2:** 30
Setup: |-18| + 12
SC-6: 18 [+/-] [g][|x|] [ENTER] 12 [+]

**A3:** 8.0
Setup: |-4.8| + |-3.2|
SC-6: 4.8 [+/-] [g][|x|] [ENTER] 3.2 [+/-] [g][|x|] [+]

**A4:** 16
Setup: |-35| - |-19|
SC-6: 35 [+/-] [g][|x|] [ENTER] 19 [+/-] [g][|x|] [-]

**A5:** |-12| > |9| (12 > 9)
Setup: 12 > 9
SC-6: Mental / Table Check

**A6:** -24
Setup: |-24|
SC-6: 24 [+/-] [g][|x|] [+/-]

**A7:** 32
Setup: 50 - |-18|
SC-6: 50 [ENTER] 18 [+/-] [g][|x|] [-]

**A8:** -|-9|, -6, 0, |-3|, |-8|
Setup: Values: 8, -6, 3, 0, -9
SC-6: Mental / Table Check

**A9:** Seagull is closer by 7 feet (25 < 32)
Setup: |-32| - 25
SC-6: 32 [+/-] [g][|x|] [ENTER] 25 [-]

**A10:** Greater debt: -$45 ($45); Greater balance: -$28
Setup: |-45|=45 > |-28|=28
SC-6: Mental / Table Check

### PRACTICE B (p.60): REAL LIFE (#1-#10)

ExB: 46 m plunge; and 50 - |-18| = 32

**B1:** |32| + |-14| = 46 meters
Setup: |32| + |-14|
SC-6: 32 [g][|x|] 14 [+/-] [g][|x|] [+]

**B2:** -$40 is greater balance; Account A owes more ($65)
Setup: |-65|
SC-6: 65 [+/-] [g][|x|]

**B3:** |-18| + |14| = 32 meters
Setup: |-18| + |14|
SC-6: 18 [+/-] [g][|x|] 14 [g][|x|] [+]

**B4:** 35 and -35 (outside minus applies after | |)
Setup: |-35|
SC-6: 35 [+/-] [g][|x|] [+/-]

**B5:** -1.5 cm (|-1.5| = 1.5 cm is smallest)
Setup: |-1.5|
SC-6: 1.5 [+/-] [g][|x|]

**B6:** 42 ÷ 6 + 11 = 18
Setup: (|-42| ÷ |-6|) + |-11|
SC-6: 42 [+/-] [g][|x|] 6 [÷/-] [g][|x|] [÷] 11 [+/-] [g][|x|] [+]

**B7:** |25| + |-140| = 165 feet
Setup: |25| + |-140|
SC-6: 25 [g][|x|] 140 [+/-] [g][|x|] [+]

**B8:** 19 and -19
Setup: |-19|
SC-6: 19 [+/-] [g][|x|]

**B9:** -|-9| (-9), -6, |5| (5), |-8| (8)
Setup: |-9|
SC-6: 9 [+/-] [g][|x|] [+/-]

**B10:** 4 5/8
Setup: (3 1/2 + 2 1/4) - 1 1/8
SC-6: 3 [.] 1 [.] 2 [ENTER] 2 [.] 1 [.] 4 [+] 1 [.] 1 [.] 8 [-]

## SEC 6.4 (pp.61-62): THE FOUR-QUADRANT COORDINATE PLANE & REFLECTIONS [COUNT-01]

Methods: W1: Plot & Count Distance to the Mirror Axis | W2: Coordinate Reflection Sign Rules

### PRACTICE A (p.61): KEY IDEAS (#1-#10)

ExA: A: Quadrant II, B: Quadrant IV, C: Quadrant III

**A1:** Quadrant IV
Setup: (+, -) -> Right, Down
SC-6: Mental / Table Check

**A2:** Quadrant III
Setup: (-, -) -> Left, Down
SC-6: Mental / Table Check

**A3:** Quadrant II
Setup: (-, +) -> Left, Up
SC-6: Mental / Table Check

**A4:** (-6, 9)
Setup: -6
SC-6: 6 [+/-] -> -6

**A5:** (6, -9)
Setup: -9
SC-6: 9 [+/-] -> -9

**A6:** (-3, 7)
Setup: 7
SC-6: 7 [+/-] [+/-] -> +7

**A7:** (8, -4)
Setup: Flip x -> (8,4); flip y -> (8,-4)
SC-6: Mental / Table Check

**A8:** (0, -9) on y-axis; (-7, 0) on x-axis
Setup: 0 sideways vs 0 up/down
SC-6: Mental / Table Check

**A9:** (-3, 3) in Quadrant II
Setup: 4 - 7; -5 + 8
SC-6: 4 [ENTER] 7 [-]; 5 [+/-] [ENTER] 8 [+]

**A10:** D(-4, -5) in Quadrant III
Setup: Match x of A (-4) & y of C (-5)
SC-6: Mental / Table Check

### PRACTICE B (p.62): REAL LIFE (#1-#10)

ExB: (5, -8) in Quadrant IV

**B1:** Cave: II, Lighthouse: I, Shipwreck: III, Pier: IV
Setup: 4
SC-6: 8 [+/-] [ENTER] 4

**B2:** (0, -7) is on y-axis; (9, 0) is on x-axis
Setup: -7
SC-6: 0 [ENTER] 7 [+/-]

**B3:** (8, -12) in Quadrant IV
Setup: -12
SC-6: 8 [ENTER] 12 [+/-]

**B4:** (9, 5) in Quadrant I
Setup: 5
SC-6: 9 [+/-] [+/-] [ENTER] 5

**B5:** Step 1: (-4, 9); Final: (4, 9) in Quad I
Setup: 9
SC-6: 4 [+/-] [+/-] [ENTER] 9 [+/-] [+/-]

**B6:** (4, -8) in Quadrant IV
Setup: -6 + 10
SC-6: 6 [+/-] [ENTER] 10 [+]

**B7:** P: Quadrant II; Q: Quadrant III
Setup: 4.5
SC-6: 2.5 [+/-] [ENTER] 4.5

**B8:** y-axis (x-coordinate changed sign, y stayed -3)
Setup: -7
SC-6: 7 [+/-]

**B9:** (-5, 5), (-5, -5), and (5, -5)
Setup: -5
SC-6: 5 [+/-]

**B10:** (x, y) is in Quad II; (-x, -y) is in Quad IV
Setup: -1
SC-6: 1 [+/-]

## SEC 6.5 (pp.63-64): DISTANCES IN THE COORDINATE PLANE [COUNT-01]

Methods: W1: Add/Subtract Absolute Values to the Axis | W2: Absolute Difference of Coordinates |x1 - x2|

### PRACTICE A (p.63): KEY IDEAS (#1-#10)

ExA: 15 units; and 8 units

**A1:** 27 units
Setup: |(-12 - 15)|
SC-6: 12 [+/-] [ENTER] 15 [-] [g][|x|]

**A2:** 11 units
Setup: |(-4 - -15)|
SC-6: 4 [+/-] [ENTER] 15 [+/-] [-] [g][|x|]

**A3:** 15 units
Setup: |(9 - -6)|
SC-6: 9 [ENTER] 6 [+/-] [-] [g][|x|]

**A4:** 9 units
Setup: |(-14 - -5)|
SC-6: 14 [+/-] [ENTER] 5 [+/-] [-] [g][|x|]

**A5:** 10 units
Setup: |(-3.5 - 6.5)|
SC-6: 3.5 [+/-] [ENTER] 6.5 [-] [g][|x|]

**A6:** 8 units
Setup: |(-4.5 - 3.5)|
SC-6: 4.5 [+/-] [ENTER] 3.5 [-] [g][|x|]

**A7:** (4, 3)
Setup: -6 + 10
SC-6: 6 [+/-] [ENTER] 10 [+]

**A8:** 13 units (They lie on opposite sides of the y-axis: |-5| + |..
Setup: |(-5 - 8)|
SC-6: 5 [+/-] [ENTER] 8 [-] [g][|x|]

**A9:** Perimeter = 34 units, Area = 72 sq units
Setup: Width 9, Height 8
SC-6: Mental / Table Check

**A10:** 20 grid units = 10 miles
Setup: 10 + 10 = 20 units × 0.5
SC-6: Mental / Table Check

### PRACTICE B (p.64): REAL LIFE (#1-#10)

ExB: Perimeter = 34 units; Area = 72 square units

**B1:** |-8| + |11| = 19 units
Setup: |(-8 - 11)|
SC-6: 8 [+/-] [ENTER] 11 [-] [g][|x|]

**B2:** 9 units = 90 meters
Setup: |(-14 - -5)|
SC-6: 14 [+/-] [ENTER] 5 [+/-] [-] [g][|x|]

**B3:** |15| + |-35| = 50 units
Setup: |(15 - -35)|
SC-6: 15 [ENTER] 35 [+/-] [-] [g][|x|]

**B4:** W = 10, H = 6, Perimeter = 32 units
Setup: |((-3 - 7)| + |(4 - -2)|) × 2
SC-6: 3 [+/-] [ENTER] 7 [-] [g][|x|] 4 [ENTER] 2 [+/-] [-] [g][|x|]..

**B5:** 60 square units
Setup: A = 10 × 6
SC-6: 10 [ENTER] 6 [f][RECT]

**B6:** 7.75 miles
Setup: 3.5 + 4.25
SC-6: 3.5 [ENTER] 4.25 [+]

**B7:** 5 3/4 units
Setup: 3 1/2 + 2 1/4
SC-6: 3 [.] 1 [.] 2 [ENTER] 2 [.] 1 [.] 4 [+]

**B8:** 7 + 10 = 17 units
Setup: (5 + 2) + (4 + 6)
SC-6: 5 [ENTER] 2 [+] 4 [ENTER] 6 [+] [+]

**B9:** (11, 4) and (-5, 4)
Setup: 3 - 8
SC-6: 3 [ENTER] 8 [+] 3 [ENTER] 8 [-]

**B10:** Base = 10, Height = 8, Area = 40 units²
Setup: A = ½((4 + 6) × (7 + 1))
SC-6: 4 [ENTER] 6 [+] 7 [ENTER] 1 [÷] [g][TRI]

# Unit 7: Equations & Inequalities — Practice A & B Complete Key

**Big Ideas Math Grade 6 — Chapter 7 • Manipulative Kits: Algebra Boxes & Stage (ALGB-01) + Counting & Array Tiles (COUNT-01)**

**Target Misconception: Students treat '=' as 'write the answer', translate '8 less than n' backwards as 8 - n, resent writing -14 under both sides of x + 14 = 39, and confuse open vs. solid inequality circles.**

Hands-On Block Remediation: Model '=' as a Balance Scale using Algebra Boxes & Stage (ALGB-01). Solve 1-step equations with inverse operations on both sides, verify your solution with the SC-6 Equality Test [≠] (which flashes NO when Left = Right!), and test inequalities on [f] [LINE] with [f] [≥] and [g] [≤] (which flash YES or NO).



**QUEST-01 Double-Points Scoring: 1 '+10' token = Textbook Method 1 (1st Way); stack a 2nd '+10' token (+20 pts) = Textbook Method 2 (2nd Way)!**

## SEC 7.1 (pp.66-67): WRITING EQUATIONS IN ONE VARIABLE & TESTING SOLUTIONS [ALGB-01]

Methods: W1: Substitute Candidates to Test Equality | W2: Balance Scale / Mental Unwrapping

### PRACTICE A (p.66): KEY IDEAS (#1-#10)

ExA: x = 9 is the solution (41 = 41)

**A1:** x = 8 (5(8)-7 = 33)
Setup: (5 × 8) - 7
SC-6: 33 [ENTER] 5 [ENTER] 8 [×] 7 [-] [≠]

**A2:** y = 15 (15÷3 + 4 = 9)
Setup: (15 ÷ 3) + 4
SC-6: 9 [ENTER] 15 [ENTER] 3 [÷] 4 [+] [≠]

**A3:** x + 14 = 52; x = 38
Setup: x + 14 = 52
SC-6: Mental / Table Check

**A4:** 6k = 84; k = 14
Setup: 6k = 84
SC-6: Mental / Table Check

**A5:** w - 9 = 17; w = 26
Setup: w - 9 = 17
SC-6: Mental / Table Check

**A6:** m / 5 = 13; m = 65
Setup: m / 5 = 13
SC-6: Mental / Table Check

**A7:** 4p + 6 = 54; p = $12
Setup: (4 × 12) + 6
SC-6: 54 [ENTER] 4 [ENTER] 12 [×] 6 [+] [≠]

**A8:** x - 12 = 20; x = 32
Setup: x - 12 = 20
SC-6: Mental / Table Check

**A9:** 10 + 3m = 28; m = 6 movies
Setup: (28 - 10) ÷ 3
SC-6: 28 [ENTER] 10 [-] 3 [÷]

**A10:** b / 3 = 14.50; b = $43.50
Setup: 14.50 × 3
SC-6: 14.50 [ENTER] 3 [×]

### PRACTICE B (p.67): REAL LIFE (#1-#10)

ExB: n - 8 = 25; n = 33 drones

**B1:** x = 8 (5·8 - 7 = 33)
Setup: (5 × 8) - 7
SC-6: 33 [ENTER] 5 [ENTER] 8 [×] 7 [-] [≠]

**B2:** 15t + 5 = 65; yes, t = 4 is a solution
Setup: (15 × 4) + 5
SC-6: 65 [ENTER] 15 [ENTER] 4 [×] 5 [+] [≠]

**B3:** k - 12 = 29; k = 41
Setup: 29 + 12
SC-6: 29 [ENTER] 12 [+]

**B4:** n ÷ 6 = 14; n = 84
Setup: 14 × 6
SC-6: 14 [ENTER] 6 [×]

**B5:** 3s = 54; s = 18 cm
Setup: 54 ÷ 3
SC-6: 54 [ENTER] 3 [÷]

**B6:** m = 15 (100 - 60 = 40)
Setup: 100 - (4 × 15)
SC-6: 40 [ENTER] 100 [ENTER] 4 [ENTER] 15 [×] [-] [≠]

**B7:** p ÷ 8 = 3; p = 24 slices
Setup: 24 ÷ 8
SC-6: 24 [ENTER] 8 [÷]

**B8:** 45 + d = 120; d = $75
Setup: 120 - 45
SC-6: 120 [ENTER] 45 [-]

**B9:** Yes (both sides equal 28)
Setup: (5 × 6) - 2
SC-6: 3 [ENTER] 6 [×] 10 [÷] 5 [ENTER] 6 [×] 2 [-] [≠]

**B10:** b - 4 = 13; b = 17 years old
Setup: 13 + 4
SC-6: 13 [ENTER] 4 [+]

## SEC 7.2 (pp.68-69): SOLVING EQUATIONS USING ADDITION OR SUBTRACTION [ALGB-01]

Methods: W1: Properties of Equality (Inverse Operations) | W2: Balance Scale / Part-Part-Whole Bar Model

### PRACTICE A (p.68): KEY IDEAS (#1-#10)

ExA: x = 45

**A1:** x = 45
Setup: 73 - 28
SC-6: 73 [ENTER] 28 [-]

**A2:** y = 92
Setup: 58 + 34
SC-6: 58 [ENTER] 34 [+]

**A3:** n = 45
Setup: 64 - 19
SC-6: 64 [ENTER] 19 [-]

**A4:** k = 76
Setup: 29 + 47
SC-6: 29 [ENTER] 47 [+]

**A5:** w = 54.4
Setup: 34.8 + 19.6
SC-6: 34.8 [ENTER] 19.6 [+]

**A6:** p = 16.65
Setup: 25.4 - 8.75
SC-6: 25.4 [ENTER] 8.75 [-]

**A7:** m = 11/24
Setup: 5/6 - 3/8
SC-6: [.] 5 [.] 6 [ENTER] [.] 3 [.] 8 [-]

**A8:** a = 4 3/4
Setup: 3 1/2 + 1 1/4
SC-6: 3 [.] 1 [.] 2 [ENTER] 1 [.] 1 [.] 4 [+]

**A9:** m - 18.50 = 26.75; m = $45.25
Setup: 26.75 + 18.50
SC-6: 26.75 [ENTER] 18.50 [+]

**A10:** h + 42.5 = 110; h = 67.5 ft
Setup: 110 - 42.5
SC-6: 110 [ENTER] 42.5 [-]

### PRACTICE B (p.69): REAL LIFE (#1-#10)

ExB: w = 54.4 g; m = 11/24

**B1:** s + 45 = 210; s = 165 points
Setup: 210 - 45
SC-6: 210 [ENTER] 45 [-]

**B2:** p - 18 = 67; p = $85
Setup: 67 + 18
SC-6: 67 [ENTER] 18 [+]

**B3:** h + 1450 = 5200; h = 3,750 feet
Setup: 5200 - 1450
SC-6: 5200 [ENTER] 1450 [-]

**B4:** f - 64.8 = 185.2; f = 250.0 grams
Setup: 185.2 + 64.8
SC-6: 185.2 [ENTER] 64.8 [+]

**B5:** t - 0.48 = 12.65; t = 13.13 seconds
Setup: 12.65 + 0.48
SC-6: 12.65 [ENTER] 0.48 [+]

**B6:** m + 3/8 = 7/8; m = 1/2 cup
Setup: 7/8 - 3/8
SC-6: [.] 7 [.] 8 [ENTER] [.] 3 [.] 8 [-]

**B7:** 2 3/4 + d = 4 1/2; d = 1 3/4 miles
Setup: 4 1/2 - 2 3/4
SC-6: 4 [.] 1 [.] 2 [ENTER] 2 [.] 3 [.] 4 [-]

**B8:** b + 38 = 125; b = 87°
Setup: 125 - 38
SC-6: 125 [ENTER] 38 [-]

**B9:** -40 + d = -115; d = -75 meters
Setup: -115 - -40
SC-6: 115 [+/-] [ENTER] 40 [+/-] [-]

**B10:** L = 44 cm; 1 length = 22 cm
Setup: (64 - 20) ÷ 2
SC-6: 64 [ENTER] 20 [-] 2 [÷]

## SEC 7.3 (pp.70-71): SOLVING EQUATIONS USING MULTIPLICATION OR DIVISION [ALGB-01]

Methods: W1: Divide/Multiply by Coefficient | W2: Multiply by the Reciprocal | W3: Equal-Groups Tape Diagram

### PRACTICE A (p.70): KEY IDEAS (#1-#10)

ExA: x = 17; k = 101.5

**A1:** x = 17
Setup: 136 ÷ 8
SC-6: 136 [ENTER] 8 [÷]

**A2:** y = 15
Setup: 180 ÷ 12
SC-6: 180 [ENTER] 12 [÷]

**A3:** m = 114
Setup: 19 × 6
SC-6: 19 [ENTER] 6 [×]

**A4:** k = 101.5
Setup: 14.5 × 7
SC-6: 14.5 [ENTER] 7 [×]

**A5:** w = 14
Setup: 63 ÷ 4.5
SC-6: 63 [ENTER] 4.5 [÷]

**A6:** x = 24
Setup: 18 ÷ 3/4
SC-6: 18 [ENTER] [.] 3 [.] 4 [÷]

**A7:** n = 40
Setup: 16 ÷ 2/5
SC-6: 16 [ENTER] [.] 2 [.] 5 [÷]

**A8:** p = 4/5 (0.8)
Setup: 2/3 ÷ 5/6
SC-6: [.] 2 [.] 3 [ENTER] [.] 5 [.] 6 [÷]

**A9:** 1.5t = 27; t = 18 minutes
Setup: 27 ÷ 1.5
SC-6: 27 [ENTER] 1.5 [÷]

**A10:** 3/5 s = 27; s = 45 students
Setup: 27 ÷ 3/5
SC-6: 27 [ENTER] [.] 3 [.] 5 [÷]

### PRACTICE B (p.71): REAL LIFE (#1-#10)

ExB: 40 songs

**B1:** 12s = 168; s = $14 per shirt
Setup: 168 ÷ 12
SC-6: 168 [ENTER] 12 [÷]

**B2:** p ÷ 9 = 3; p = 27 slices
Setup: 3 × 9
SC-6: 3 [ENTER] 9 [×]

**B3:** 65t = 325; t = 5 hours
Setup: 325 ÷ 65
SC-6: 325 [ENTER] 65 [÷]

**B4:** 2.5p = 16.25; p = $6.50 per pound
Setup: 16.25 ÷ 2.5
SC-6: 16.25 [ENTER] 2.5 [÷]

**B5:** s ÷ 4 = 9; s = $36
Setup: 9 × 4
SC-6: 9 [ENTER] 4 [×]

**B6:** (3/4)c = 21; c = 28 students
Setup: 21 ÷ 3/4
SC-6: 21 [ENTER] [.] 3 [.] 4 [÷]

**B7:** (5/6)p = 150; p = $180
Setup: 150 ÷ 5/6
SC-6: 150 [ENTER] [.] 5 [.] 6 [÷]

**B8:** 13.5w = 108; w = 8 feet
Setup: 108 ÷ 13.5
SC-6: 108 [ENTER] 13.5 [÷]

**B9:** h ÷ 15 = 3.2; h = 48 feet
Setup: 3.2 × 15
SC-6: 3.2 [ENTER] 15 [×]

**B10:** 0.40n = 56; n = 140
Setup: 56 ÷ 0.40
SC-6: 56 [ENTER] 0.40 [÷]

## SEC 7.4 (pp.72-73): WRITING AND GRAPHING INEQUALITIES [ALGB-01]

Methods: W1: Inverse Property of Inequality & Number Line Graph | W2: Boundary & Test-Point Substitution

### PRACTICE A (p.72): KEY IDEAS (#1-#10)

ExA: x ≥ 19 (Solid dot at 19, arrow right)

**A1:** x > 13 (Open circle at 13, arrow right)
Setup: 21 - 8
SC-6: 21 [ENTER] 8 [-]

**A2:** y ≤ 24 (Solid dot at 24, arrow left)
Setup: 15 + 9
SC-6: 15 [ENTER] 9 [+]

**A3:** n ≥ 13 (Solid dot at 13, arrow right)
Setup: 78 ÷ 6
SC-6: 78 [ENTER] 6 [÷]

**A4:** m < 36 (Open circle at 36, arrow left)
Setup: 9 × 4
SC-6: 9 [ENTER] 4 [×]

**A5:** h ≥ 48
Setup: 'At least' = ≥
SC-6: Mental / Table Check

**A6:** w ≤ 2,000
Setup: 'At most' = ≤
SC-6: Mental / Table Check

**A7:** Only 6 (At n = 5, 23 > 23 is false)
Setup: Compare 23 > ((4 × 6) + 3)
SC-6: 23 [ENTER] 4 [ENTER] 6 [×] 3 [+] [f] [>]

**A8:** 5 and 6
Setup: (4 × 5) - 3
SC-6: 17 [ENTER] 4 [ENTER] 5 [×] 3 [-] [f][≥]

**A9:** 8b ≤ 50; at most 6 books ($2 left over)
Setup: 50 ÷ 8 (Q & R)
SC-6: 50 [ENTER] 8 [g] [÷R]

**A10:** p + 340 ≥ 500; p ≥ 160 points
Setup: 500 - 340
SC-6: 500 [ENTER] 340 [-]

### PRACTICE B (p.73): REAL LIFE (#1-#10)

ExB: g ≤ 15 (at most 15 gold bars)

**B1:** h ≥ 54 (solid dot at 54, shade right); at least 5 in
Setup: 54 - 49
SC-6: 54 [ENTER] 49 [-]

**B2:** p + 5 ≤ 12 -> p ≤ 7 passengers
Setup: 12 - 5
SC-6: 12 [ENTER] 5 [-]

**B3:** 22.50 + x ≥ 35 -> x ≥ $12.50
Setup: 35 - 22.50
SC-6: 35 [ENTER] 22.50 [-]

**B4:** 3.50g ≤ 28 -> g ≤ 8 games
Setup: 28 ÷ 3.50
SC-6: 28 [ENTER] 3.50 [÷]

**B5:** 15r > 240 -> r > 16 (at least 17 rows)
Setup: 240 ÷ 15
SC-6: 240 [ENTER] 15 [÷]

**B6:** b - 25 < 15 -> b < 40%
Setup: 15 + 25
SC-6: 15 [ENTER] 25 [+]

**B7:** 8 and 10
Setup: (5 × 8) - 10
SC-6: 30 [ENTER] 5 [ENTER] 8 [×] 10 [-] [f][≥]

**B8:** 2c - 30 ≥ 70 -> c ≥ 50 cupcakes
Setup: (70 + 30) ÷ 2
SC-6: 70 [ENTER] 30 [+] 2 [÷]

**B9:** s ≤ 20 (solid dot at 20, shade left down to 0)
Setup: Compare 20 ≤ 20
SC-6: 20 [ENTER] 20 [g][≤]

**B10:** 45c + 170 ≤ 800 -> c ≤ 14 crates
Setup: (800 - 170) ÷ 45
SC-6: 800 [ENTER] 170 [-] 45 [÷]

## SEC 7.5 (pp.74-75): WRITING EQUATIONS IN TWO VARIABLES (Y = KX AND Y = X + B) [COUNT-01]

Methods: W1: Test Constant Ratio y/x vs. Constant Difference y - x | W2: Input-Output Table & Coordinate Graph

### PRACTICE A (p.74): KEY IDEAS (#1-#10)

ExA: Table A: y = 7x; Table B: y = x - 4

**A1:** y = 8x; y = 80
Setup: 16 ÷ 2; 8×10
SC-6: 16 [ENTER] 2 [÷] -> 8; 8×10

**A2:** y = x + 8; y = 28
Setup: 9 - 1; 20+8
SC-6: 9 [ENTER] 1 [-] -> +8; 20+8

**A3:** y = 7x; y = 84
Setup: 14 ÷ 2; 7×12
SC-6: 14 [ENTER] 2 [÷] -> 7; 7×12

**A4:** y = x - 4; y = 26
Setup: 30 - 4
SC-6: 30 [ENTER] 4 [-]

**A5:** (1, 6), (2, 12), (3, 18), (4, 24)
Setup: Repeat(+6)
SC-6: 6 [g] [REP] [+] [+] [+]

**A6:** Independent: t; Dependent: d
Setup: d depends on t
SC-6: Mental / Table Check

**A7:** 292.5 miles
Setup: 65 × 4.5
SC-6: 65 [ENTER] 4.5 [×]

**A8:** y = 4x passes through (0, 0); y = x + 4 crosses at (0, 4)
Setup: Test x = 0 in both
SC-6: Mental / Table Check

**A9:** 8 hours
Setup: 100 ÷ 12.50
SC-6: 100 [ENTER] 12.50 [÷]

**A10:** c = 15m + 25; $205
Setup: (15 × 12) + 25
SC-6: 15 [ENTER] 12 [×] 25 [+]

### PRACTICE B (p.75): REAL LIFE (#1-#10)

ExB: p = 12.50h; $81.25

**B1:** y = x + 5 is additive; y = 8m is multiplicative
Setup: 8 × 1
SC-6: 8 [ENTER] 1 [×]

**B2:** g = 18h (h = indep, g = dep); 81 grams
Setup: 18 × 4.5
SC-6: 18 [ENTER] 4.5 [×]

**B3:** m = z + 4; m = 12, 16, 19
Setup: 8 + 4
SC-6: 8 [ENTER] 4 [+]

**B4:** s = 15p; p = 12 sheets
Setup: 180 ÷ 15
SC-6: 180 [ENTER] 15 [÷]

**B5:** y = x - 6; $18, $29, and $44
Setup: 24 - 6
SC-6: 24 [ENTER] 6 [-]

**B6:** y = 3.5x; 140 wheel turns
Setup: 3.5 × 40
SC-6: 3.5 [ENTER] 40 [×]

**B7:** y = 9x; at x = 20, y = 180
Setup: (27 ÷ 3) × 20
SC-6: 27 [ENTER] 3 [÷] 20 [×]

**B8:** c = 20g + 7; $127
Setup: (20 × 6) + 7
SC-6: 20 [ENTER] 6 [×] 7 [+]

**B9:** Multiplicative y = kx (when x = 0, y = k·0 = 0)
Setup: 0 × 7
SC-6: 0 [ENTER] 7 [×]

**B10:** d = 0.2s; 3 miles away
Setup: 0.2 × 15
SC-6: 0.2 [ENTER] 15 [×]

# Unit 8: Surface Area & Volume of 3D Solids — Practice A & B Complete Key

**Big Ideas Math Grade 6 — Chapter 8 • Manipulative Kits: Counting & Array Tiles (COUNT-01) + Gridfinity 3x1 Storage Boxes**

**Target Misconception: Students drop one of the 6 faces when finding rectangular prism surface area, tangle prism and pyramid formulas, and confuse 2D Surface Area (cardboard outside) with 3D Volume (cubes inside).**

Hands-On Block Remediation: Have students touch the 3 pairs of faces on their physical Gridfinity box while logging each pair into [Σ+] (n=1, 2, 3), use [f] [PRISM] ([f][9]) to multiply 3 stack floors (l, w, h) at once for volume, and unlock [f] [CIRC], [g] [CONE], and [g] [HYP].



**QUEST-01 Double-Points Scoring: 1 '+10' token = Textbook Method 1 (1st Way); stack a 2nd '+10' token (+20 pts) = Textbook Method 2 (2nd Way)!**

## SEC 8.1 (pp.77-78): THREE-DIMENSIONAL FIGURES, NETS & RECTANGULAR PRISMS [COUNT-01]

Methods: W1: Unfold into a 6-Face 2D Net | W2: Opposite Face Pairs SA = 2(lw + lh + wh) | W3: Lateral Strip Ph + 2B

### PRACTICE A (p.77): KEY IDEAS (#1-#10) — ExA: 158 cm²

**A1:** 96 cm²
Setup: 4² × 6
SC-6: 4 [x²] 6 [×]

**A2:** 486 in.²
Setup: 9² × 6
SC-6: 9 [x²] 6 [×]

**A3:** 108 cm²
Setup: 2(24)+2(18)+2(12)
SC-6: Mental / Table Check

**A4:** 160 in.²
Setup: 2(50)+2(20)+2(10)
SC-6: Mental / Table Check

**A5:** 392 ft²
Setup: 2(96)+2(60)+2(40)
SC-6: Mental / Table Check

**A6:** 85 m²
Setup: 2(20)+2(12.5)+2(10)
SC-6: Mental / Table Check

**A7:** One face = 25 cm²; s = 5 cm
Setup: √((150 ÷ 6))
SC-6: 150 [ENTER] 6 [÷] [f][√x]

**A8:** 94 cm² (He only summed 3 faces instead of doubling for all 6..
Setup: 47 × 2
SC-6: 47 [ENTER] 2 [×]

**A9:** 2,100 cm²
Setup: 600 + 2(450) + 2(300)
SC-6: Mental / Table Check

**A10:** 500 in.³ needed; Yes, with 50 in.³ to spare
Setup: 2(150)+2(60)+2(40)
SC-6: Mental / Table Check

### PRACTICE B (p.78): REAL LIFE (#1-#10) — ExB: Cube = 294 in²; Open-Top Bin = 260 cm²

**B1:** 2(160 + 600 + 240) = 2,000 cm²
Setup: (((20 × 8) + (20 × 30)) + (8 × 30)) × 2
SC-6: 20 [ENTER] 8 [×] 20 [ENTER] 30 [×] [+] 8 [ENTER] 30 [×] [+] 2..

**B2:** 1 face = 81 in²; Total SA = 486 in²
Setup: 9² × 6
SC-6: 9 [x²] 6 [×]

**B3:** 288 + 720 + 360 = 1,368 in²
Setup: ((24 × 12) + ((24 × 15) × 2)) + ((12 × 15) × 2)
SC-6: 24 [ENTER] 12 [×] 24 [ENTER] 15 [×] 2 [×] [+] 12 [ENTER] 15 [×..

**B4:** SA = 500 in²; yes (100 in² to spare)
Setup: (((15 × 10) + (15 × 4)) + (10 × 4)) × 2
SC-6: 15 [ENTER] 10 [×] 15 [ENTER] 4 [×] [+] 10 [ENTER] 4 [×] [+] 2..

**B5:** 120 + 192 + 160 = 472 ft²
Setup: ((12 × 10) + ((12 × 8) × 2)) + ((10 × 8) × 2)
SC-6: 12 [ENTER] 10 [×] 12 [ENTER] 8 [×] 2 [×] [+] 10 [ENTER] 8 [×]..

**B6:** 1 face = 25 cm²; edge = 5 cm
Setup: √((150 ÷ 6))
SC-6: 150 [ENTER] 6 [÷] [f][√x]

**B7:** 32 + 8 = 40 ft²
Setup: ((4 × 2) × 4) + ((2 × 2) × 2)
SC-6: 4 [ENTER] 2 [×] 4 [×] 2 [ENTER] 2 [×] 2 [×] [+]

**B8:** 2(24 + 60 + 40) = 248 cm²
Setup: (((6 × 4) + (6 × 10)) + (4 × 10)) × 2
SC-6: 6 [ENTER] 4 [×] 6 [ENTER] 10 [×] [+] 4 [ENTER] 10 [×] [+] 2 [×]

**B9:** 54 cm² vs. 216 cm² (4 times as great)
Setup: (6² × 6) ÷ 54
SC-6: 6 [x²] 6 [×] 54 [÷]

**B10:** 160 cm²
Setup: 4² × 10
SC-6: 4 [x²] 10 [×]

## SEC 8.2 (pp.79-80): SURFACE AREAS OF TRIANGULAR PRISMS AND PYRAMIDS [COUNT-01]

Methods: W1: Add All Individual Faces on the 2D Net | W2: Base Area + Combined Lateral Area

### PRACTICE A (p.79): KEY IDEAS (#1-#10) — ExA: 340 m²

**A1:** 96 cm²
Setup: 6² + (½(6×5) × 4)
SC-6: 6 [x²] 6 [ENTER] 5 [g][TRI] 4 [×] [+]

**A2:** 224 in.²
Setup: 8² + (½(8×10) × 4)
SC-6: 8 [x²] 8 [ENTER] 10 [g][TRI] 4 [×] [+]

**A3:** 360 ft²
Setup: 12² + (½(12×9) × 4)
SC-6: 12 [x²] 12 [ENTER] 9 [g][TRI] 4 [×] [+]

**A4:** 62.4 cm²
Setup: ½(6×5.2) × 4
SC-6: 6 [ENTER] 5.2 [g][TRI] 4 [×]

**A5:** 132 cm²
Setup: ½(4×3) × 2
SC-6: 4 [ENTER] 3 [g][TRI] 2 [×] (12) + 12×10 (120)

**A6:** 150 ft²
Setup: ½(8×3) × 2
SC-6: 8 [ENTER] 3 [g][TRI] 2 [×] (24) + 18×7 (126)

**A7:** 110.88 cm²
Setup: ½(8×6.93) × 4
SC-6: 8 [ENTER] 6.93 [g][TRI] 4 [×]

**A8:** Triangular prism: 2 triangles + 3 rectangles; Square pyramid..
Setup: Count net faces
SC-6: Mental / Table Check

**A9:** 152 ft²
Setup: 24 + 48 + 80
SC-6: Mental / Table Check

**A10:** 108 ft²
Setup: ½(9×6) × 4
SC-6: 9 [ENTER] 6 [g][TRI] 4 [×]

### PRACTICE B (p.80): REAL LIFE (#1-#10) — ExB: 288 square feet

**B1:** 24 + 80 + 48 = 152 ft²
Setup: ((½(6×4) × 2) + ((5×8) × 2)) + (6×8)
SC-6: 6 [ENTER] 4 [g][TRI] 2 [×] 5 [ENTER] 8 [f][RECT] 2 [×] [+] 6 [..

**B2:** 144 + 240 = 384 cm²
Setup: 12² + (½(12×10) × 4)
SC-6: 12 [x²] 12 [ENTER] 10 [g][TRI] 4 [×] [+]

**B3:** 96 ft²
Setup: ½(8×6) × 4
SC-6: 8 [ENTER] 6 [g][TRI] 4 [×]

**B4:** 13.8 + 240 = 253.8 cm²
Setup: (6.9 × 2) + ((4×20) × 3)
SC-6: 6.9 [ENTER] 2 [×] 4 [ENTER] 20 [f][RECT] 3 [×] [+]

**B5:** 48 + 120 = 168 in²
Setup: (½(8×6) × 2) + (((8 + 6) + 10) × 5)
SC-6: 8 [ENTER] 6 [g][TRI] 2 [×] 8 [ENTER] 6 [+] 10 [+] 5 [×] [+]

**B6:** 15.6 cm²
Setup: ½(3×2.6) × 4
SC-6: 3 [ENTER] 2.6 [g][TRI] 4 [×]

**B7:** 96 + 600 = 696 ft²
Setup: (½(16×6) × 2) + ((10×30) × 2)
SC-6: 16 [ENTER] 6 [g][TRI] 2 [×] 10 [ENTER] 30 [f][RECT] 2 [×] [+]

**B8:** 15 in² per face; slant height = 5 in
Setup: (96 - 36) ÷ 4
SC-6: 96 [ENTER] 36 [-] 4 [÷]

**B9:** 96 + 480 = 576 cm²
Setup: (½(12×8) × 2) + (((10 + 10) + 12) × 15)
SC-6: 12 [ENTER] 8 [g][TRI] 2 [×] 10 [ENTER] 10 [+] 12 [+] 15 [×] [+]

**B10:** Pyramid A (260 cm² > 224 cm²)
Setup: 8² + (½(8×10) × 4)
SC-6: 10 [x²] 10 [ENTER] 8 [g][TRI] 4 [×] [+] 8 [x²] 8 [ENTER] 10 [g..

## SEC 8.3 (pp.81-82): VOLUMES OF RECTANGULAR PRISMS [COUNT-01]

Methods: W1: Base Area × Height (V = Bh) | W2: Multiply Length × Width × Height (Fractions or Decimals)

### PRACTICE A (p.81): KEY IDEAS (#1-#10) — ExA: Crate = 480 in.³; Aquarium = 10 1/2 ft³

**A1:** 3,024 cm³
Setup: Prism(18×6×28)
SC-6: 18 [ENTER] 6 [ENTER] 28 [f][PRISM]

**A2:** 125 in.³
Setup: 5³
SC-6: 5 [x³]

**A3:** 500 m³
Setup: Prism(25×10×2)
SC-6: 25 [ENTER] 10 [ENTER] 2 [f][PRISM]

**A4:** 224 m³
Setup: 32 × 7
SC-6: 32 [ENTER] 7 [×]

**A5:** 10 1/2 ft³ (10.5)
Setup: Prism(3.5×1.5×2)
SC-6: 3.5 [ENTER] 1.5 [ENTER] 2 [f][PRISM]

**A6:** 27 in.³
Setup: Prism(4.5×3×2)
SC-6: 4.5 [ENTER] 3 [ENTER] 2 [f][PRISM]

**A7:** 6 in.
Setup: 270 ÷ (9 × 5)
SC-6: 270 [ENTER] 9 [ENTER] 5 [×] [+]

**A8:** 8 half-inch cubes
Setup: 2³
SC-6: 2 [x³]

**A9:** 8 inches
Setup: 360 ÷ 45
SC-6: 360 [ENTER] 45 [÷]

**A10:** 216 half-inch cubes
Setup: 27 × 2³
SC-6: 27 [ENTER] 2 [x³] [×]

### PRACTICE B (p.82): REAL LIFE (#1-#10) — ExB: 8 inches deep

**B1:** 500 m³
Setup: Prism(25×10×2)
SC-6: 25 [ENTER] 10 [ENTER] 2 [f][PRISM]

**B2:** 48 ft³
Setup: Prism(8×4×1.5)
SC-6: 8 [ENTER] 4 [ENTER] 1.5 [f][PRISM]

**B3:** 15,625 cm³
Setup: 25³
SC-6: 25 [x³]

**B4:** 65 in³
Setup: Prism(6.5×4×2.5)
SC-6: 6.5 [ENTER] 4 [ENTER] 2.5 [f][PRISM]

**B5:** 8 per in³; 96 half-inch cubes
Setup: 12 × 8
SC-6: 12 [ENTER] 8 [×]

**B6:** 3 cm rise
Setup: 120 ÷ 40
SC-6: 120 [ENTER] 40 [÷]

**B7:** 77 ft³
Setup: Prism(5.5×4×3.5)
SC-6: 5.5 [ENTER] 4 [ENTER] 3.5 [f][PRISM]

**B8:** 36 ft³
Setup: (12 × 9) ÷ 3
SC-6: 12 [ENTER] 9 [×] 3 [÷]

**B9:** 8 inches
Setup: 360 ÷ 45
SC-6: 360 [ENTER] 45 [÷]

**B10:** 4,608 in³ ≈ 19.95 (about 20) gallons
Setup: Prism(24×12×16) ÷ 231
SC-6: 24 [ENTER] 12 [ENTER] 16 [f][PRISM] 231 [÷]

## SEC 8.4 (pp.83-84): CIRCLES, CONES, AND THE PYTHAGOREAN THEOREM [COUNT-01]

Methods: W1: Exact Form in Terms of π & a² + b² = c² | W2: Decimal Approximation (π ≈ 3.14) & Scaled Triples

### PRACTICE A (p.83): KEY IDEAS (#1-#10) — ExA: Pizza = 113.10 in.²; Ramp diagonal = 13 ft

**A1:** 78.54 cm²
Setup: π × 5²
SC-6: 5 [f][CIRC]

**A2:** 706.86 cm²
Setup: π × 15²
SC-6: 15 [f][CIRC]

**A3:** 5 cm
Setup: √(3² + 4²)
SC-6: 3 [ENTER] 4 [g][HYP]

**A4:** 10 meters
Setup: √(6² + 8²)
SC-6: 6 [ENTER] 8 [g][HYP]

**A5:** 13 ft
Setup: √(12² + 5²)
SC-6: 12 [ENTER] 5 [g][HYP]

**A6:** r = 8 m; Area = 201.06 m²
Setup: π × (16 ÷ 2)²
SC-6: 16 [ENTER] 2 [÷] [f][CIRC]

**A7:** 18.85 m²
Setup: ⅓π(3²)(2)
SC-6: 3 [ENTER] 2 [g][CONE]

**A8:** 188.50 in.³
Setup: ⅓π(6²)(5)
SC-6: 6 [ENTER] 5 [g][CONE]

**A9:** Cylinder = 282.74 cm³; Cone = 94.25 cm³
Setup: ⅓π(3²)(10)
SC-6: 3 [ENTER] 10 [g][CONE]

**A10:** 40 inches
Setup: √(32² + 24²)
SC-6: 32 [ENTER] 24 [g][HYP]

### PRACTICE B (p.84): REAL LIFE (#1-#10) — ExB: 94.25 cm³ (exact: 30π cm³)

**B1:** 153.94 ft²
Setup: π × 7²
SC-6: 7 [f][CIRC]

**B2:** r = 20 m; Area ≈ 1,256.64 m²
Setup: π × 20²
SC-6: 20 [f][CIRC]

**B3:** 87.96 inches
Setup: (2 × π) × 14
SC-6: 2 [ENTER] [π] [×] 14 [×]

**B4:** 25 meters
Setup: √(15² + 20²)
SC-6: 15 [ENTER] 20 [g][HYP]

**B5:** 20 inches
Setup: √(16² + 12²)
SC-6: 16 [ENTER] 12 [g][HYP]

**B6:** 150.80 cm³ (48π)
Setup: ⅓π(4²)(9)
SC-6: 4 [ENTER] 9 [g][CONE]

**B7:** 565.49 in³ (180π)
Setup: ⅓π(6²)(15)
SC-6: 6 [ENTER] 15 [g][CONE]

**B8:** One 16-in pizza (201.06 in² > 157.08 in²)
Setup: π × 5² × 2
SC-6: 8 [f][CIRC] 5 [f][CIRC] 2 [×]

**B9:** 56.55 ft²
Setup: π × 6² ÷ 2
SC-6: 6 [f][CIRC] 2 [÷]

**B10:** 17 feet
Setup: √(8² + 15²)
SC-6: 8 [ENTER] 15 [g][HYP]

# Unit 9: Statistical Measures & Data Displays — Practice A & B Complete Key

**Big Ideas Math Grade 6 — Chapters 9 & 10 (Statistical Measures and Data Displays) • Manipulative Kits: Counting & Array Tiles (COUNT-01) + Probability Dice Studio (PROB-01)**

**Target Misconception: Students forget to sort data before finding Median/Quartiles, lose their place adding long lists for Mean, and dread Mean Absolute Deviation (MAD) because doing it by hand takes 20 minutes of subtraction.**

Hands-On Block Remediation: Show that Mean is physically leveling unequal cube towers to the same height. Then show that entering a list ONCE with [Σ+] unlocks Mean ([g] [x-bar]), Median ([f] [MED]), Mode ([g] [MODE]), Range ([RNG]), Quartiles ([f] [1Q], [g] [3Q], [IQR]), and MAD ([g] [MAD]) in 1 click each!



**QUEST-01 Double-Points Scoring: 1 '+10' token = Textbook Method 1 (1st Way); stack a 2nd '+10' token (+20 pts) = Textbook Method 2 (2nd Way)!**

## SEC 9.1 (pp.86-87): STATISTICAL QUESTIONS, DOT PLOTS, AND THE MEAN [COUNT-01]

Methods: W1: Mean Formula (Sum ÷ Count) | W2: Fair-Share Block Leveling | W3: Balance Point Deviations

### PRACTICE A (p.86): KEY IDEAS (#1-#10)

ExA: Sum = 105, Count = 5, Mean = 21

**A1:** Sum = 20, Mean = 5
Setup: Mean(2, 4, 6, 8)
SC-6: [g][CLΣ] 2 [Σ+] 4 [Σ+] 6 [Σ+] 8 [Σ+] [g][x-bar]

**A2:** Sum = 70, Mean = 14
Setup: Sort/Load 5 values -> Stats
SC-6: Load 5 nums into [Σ+] -> [g][x-bar]

**A3:** 156 cm
Setup: Mean(150, 162, 156)
SC-6: 150 [Σ+] 162 [Σ+] 156 [Σ+] [g][x-bar]

**A4:** Mean = 4.0 g
Setup: Load data -> Stats
SC-6: Load into [Σ+] -> [g][x-bar]

**A5:** Sum = 55, Mean = 18.33
Setup: Mean(35, 25)
SC-6: ... 35 [f][Σ-] 25 [Σ+] [g][x-bar]

**A6:** 13.33 (80 ÷ 6)
Setup: Mean(10, 4, 20, 2)
SC-6: 10 [ENTER] 4 [FRQ] 20 [ENTER] 2 [FRQ] [g][x-bar]

**A7:** 70°F
Setup: Sort/Load 6 values -> Stats
SC-6: Load 6 temps into [Σ+] -> [g][x-bar]

**A8:** 60 total blocks
Setup: 5 × 12
SC-6: 5 [ENTER] 12 [×]

**A9:** 95
Setup: 360 - 265
SC-6: 360 [ENTER] 265 [-]

**A10:** 17 points
Setup: (90 + 29) ÷ 7
SC-6: 90 [ENTER] 29 [+] 7 [÷]

### PRACTICE B (p.87): REAL LIFE (#1-#10)

ExB: 94

**B1:** Sum = 85; Mean = 17 points
Setup: Mean(14, 18, 12, 22, 19)
SC-6: [g][CLΣ] 14 [Σ+] 18 [Σ+] 12 [Σ+] 22 [Σ+] 19 [Σ+] [g][x-bar]

**B2:** 42 grams
Setup: Mean(41, 43, 42, 40, 44...)
SC-6: [g][CLΣ] 41 [Σ+] 43 [Σ+] 42 [Σ+] 40 [Σ+] 44 [Σ+] 42 [Σ+] [g][x-bar]

**B3:** 89 ÷ 10 = 8.9
Setup: (((3 × 8) + (5 × 9)) + (2 × 10)) ÷ 10
SC-6: 3 [ENTER] 8 [×] 5 [ENTER] 9 [×] [+] 2 [ENTER] 10 [×] [+] 10 [÷]

**B4:** 91
Setup: (5 × 85) - (((80 + 88) + 82) + 84)
SC-6: 5 [ENTER] 85 [×] 80 [ENTER] 88 [+] 82 [+] 84 [+] [-]

**B5:** Total = 36,000; Mean = 9,000 steps
Setup: (((8200 + 9500) + 7800) + 10500) ÷ 4
SC-6: 8200 [ENTER] 9500 [+] 7800 [+] 10500 [+] 4 [÷]

**B6:** 82.5 -> 90.0 (+7.5 points)
Setup: ((88 + 92) + 90) ÷ 3
SC-6: 88 [ENTER] 92 [+] 90 [+] 3 [÷]

**B7:** 12 books per stack
Setup: (((6 + 10) + 14) + 18) ÷ 4
SC-6: 6 [ENTER] 10 [+] 14 [+] 18 [+] 4 [÷]

**B8:** 4.8 g
Setup: ((((4.2 + 5.1) + 4.8) + 5.5) + 4.4) ÷ 5
SC-6: 4.2 [ENTER] 5.1 [+] 4.8 [+] 5.5 [+] 4.4 [+] 5 [÷]

**B9:** Reading time varies from student to student
Setup: 60 × 1
SC-6: 60 [ENTER] 1 [×]

**B10:** 260 ÷ 10 = 26 points
Setup: (80 + 180) ÷ 10
SC-6: 80 [ENTER] 180 [+] 10 [÷]

## SEC 9.2 (pp.88-89): MEASURES OF CENTER: MEDIAN, MODE, AND RANGE [PROB-01]

Methods: W1: Ordered List (Least to Greatest) | W2: Dot Plot / Frequency Table Analysis

### PRACTICE A (p.88): KEY IDEAS (#1-#10)

ExA: Mean = 7.875, Median = 7.5, Mode = 7, Range = 4

**A1:** Mode = 7
Setup: Sort/Load 8 values -> Mode
SC-6: Load 8 nums -> [g][MODE]

**A2:** Median = 14, Mode = 12, Range = 8
Setup: Sort/Load 6 values -> Med/Mode/Range
SC-6: Load 6 nums -> [f][MED], [g][MODE], [RNG]

**A3:** Mean = 9, Median = 8, Mode = 8, Range = 9
Setup: Sort/Load 5 values -> Stats
SC-6: Load 5 nums into [Σ+]

**A4:** Min = 62°F, Max = 85°F, Range = 23°F
Setup: Sort/Load 7 values -> Range
SC-6: Load 7 nums -> [f][MIN], [g][MAX], [RNG]

**A5:** 75°F
Setup: Sort/Load 6 values -> Med
SC-6: Load 6 nums -> [f][MED]

**A6:** Mean = $29.17, Median = $17.50 (Median is best)
Setup: Sort/Load 6 values -> Mean/Med
SC-6: Load 6 nums -> [g][x-bar], [f][MED]

**A7:** Mean increases from 14 to 30.8; Median stays 14
Setup: Compare both sets
SC-6: Mental / Table Check

**A8:** 95
Setup: 64 + 31
SC-6: 64 [ENTER] 31 [+]

**A9:** Median = 14, Range = 13
Setup: Sorted: 8, 11, 14, 16, 21
SC-6: Mental / Table Check

**A10:** Mean = $370k, Median = $220k (Use Median due to the $980k ou..
Setup: Sort/Load 5 values -> Mean/Med
SC-6: Load 5 nums -> [g][x-bar], [f][MED]

### PRACTICE B (p.89): REAL LIFE (#1-#10)

ExB: Mean = $29.17, Median = $17.50, Mode = $15 (Median is best)

**B1:** Median = 62, Mode = 62, Range = 22
Setup: Med/Mode/Range(55, 62, 48, 70, 62...)
SC-6: [g][CLΣ] 55 [Σ+] 62 [Σ+] 48 [Σ+] 70 [Σ+] 62 [Σ+] 50 [Σ+] 66 [Σ+] [f][MED] [g][MODE] [RNG]

**B2:** Median = 12.5, Mode = 12, Range = 5
Setup: Med(11, 13, 12, 14, 12...)
SC-6: [g][CLΣ] 11 [Σ+] 13 [Σ+] 12 [Σ+] 14 [Σ+] 12 [Σ+] 16 [Σ+] [f][MED]

**B3:** Mean=$40, Med=$25, Mode=$20; Median (outlier $105 skews mean)
Setup: Mean/Med(20, 25, 20, 30, 105)
SC-6: [g][CLΣ] 20 [Σ+] 25 [Σ+] 20 [Σ+] 30 [Σ+] 105 [Σ+] [g][x-bar] [f][MED]

**B4:** Modes = 6 and 8 (bimodal); Median = 7
Setup: 8
SC-6: 6 [ENTER] 8

**B5:** Median = 11.8 s, Range = 1.2 s, No mode
Setup: 12.4 - 11.2
SC-6: 12.4 [ENTER] 11.2 [-]

**B6:** Min = 64, Max = 81, Range = 17°F, Median = 72°F
Setup: Med/Range(72, 68, 75, 81, 64...)
SC-6: [g][CLΣ] 72 [Σ+] 68 [Σ+] 75 [Σ+] 81 [Σ+] 64 [Σ+] 70 [Σ+] 74 [Σ+] [f][MED] [RNG]

**B7:** Mean = 11, Median = 11, Mode = 12, Range = 7
Setup: Mean/Med/Mode/Range(8, 12, 9, 15, 12...)
SC-6: [g][CLΣ] 8 [Σ+] 12 [Σ+] 9 [Σ+] 15 [Σ+] 12 [Σ+] 10 [Σ+] [g][x-bar] [f][MED] [g][MODE] [RNG]

**B8:** Example: 8, 8, 10, 12, 15
Setup: 15 - 8
SC-6: 15 [ENTER] 8 [-]

**B9:** Median increases by 5 (to 85); Range stays 20
Setup: 95 - 75
SC-6: 95 [ENTER] 75 [-]

**B10:** Mean = 11.75, Median = 11.5, Mode = 10, Range = 6
Setup: Mean/Med(9, 12, 10, 14, 11...)
SC-6: [g][CLΣ] 9 [Σ+] 12 [Σ+] 10 [Σ+] 14 [Σ+] 11 [Σ+] 15 [Σ+] 10 [Σ+] 13 [Σ+] [g][x-bar] [f][MED]

## SEC 9.3 (pp.90-91): QUARTILES, INTERQUARTILE RANGE (IQR) & BOX PLOTS [FRAC-01]

Methods: W1: Split Ordered Data into Lower & Upper Halves | W2: Box-and-Whisker Plot & 25% Quartile Sections

### PRACTICE A (p.90): KEY IDEAS (#1-#10)

ExA: Min = 68, Q1 = 73.5, Med = 82, Q3 = 90, Max = 96; IQR = 16.5

**A1:** Min = 10, Q1 = 15, Med = 30, Q3 = 45, Max = 50, IQR = 30
Setup: Sort/Load 5 values -> Q1/Q3/IQR
SC-6: Load 5 nums -> [f][1Q], [g][3Q], [IQR]

**A2:** Q1 = 6, Med = 10, Q3 = 14, IQR = 8
Setup: Sort/Load 7 values -> Q1/Q3/IQR
SC-6: Load 7 nums -> [f][1Q], [g][3Q], [IQR]

**A3:** 34
Setup: 58 - 24
SC-6: 58 [ENTER] 24 [-]

**A4:** 75%
Setup: 3 quarters = 3 × 25%
SC-6: Mental / Table Check

**A5:** Q1 = 73.5, Med = 82, Q3 = 90, IQR = 16.5
Setup: Sort/Load 8 values -> Stats
SC-6: Load 8 nums into [Σ+]

**A6:** Q1 = 46.5, Med = 51, Q3 = 56.5, IQR = 10
Setup: Sort/Load 8 values -> Stats
SC-6: Load 8 nums into [Σ+]

**A7:** Player A (smaller IQR means the middle 50% of scores are clo..
Setup: Compare box widths
SC-6: Mental / Table Check

**A8:** 90 (Any score > 90 is an outlier)
Setup: 60 + (1.5 × 20)
SC-6: 60 [ENTER] 1.5 [ENTER] 20 [×] [+]

**A9:** Q1 = 15, Med = 22, Q3 = 28, IQR = 13
Setup: Sort/Load 7 values -> Q1/Q3/IQR
SC-6: Load 7 nums -> [IQR]

**A10:** About 100 students (50% of 200)
Setup: 50% of 200
SC-6: 200 [ENTER] 50 [%]

### PRACTICE B (p.91): REAL LIFE (#1-#10)

ExB: IQR = 16; 20 students (50%)

**B1:** Min=40, Q1=45, Med=55, Q3=65, Max=70; IQR = 20
Setup: Q1/Q3/IQR(40, 45, 50, 55, 60...)
SC-6: [g][CLΣ] 40 [Σ+] 45 [Σ+] 50 [Σ+] 55 [Σ+] 60 [Σ+] 65 [Σ+] 70 [Σ+] [IQR]

**B2:** Q1 = 7.5, Med = 10, Q3 = 13, IQR = 5.5
Setup: Q1/Q3/IQR(6, 7, 8, 9, 11...)
SC-6: [g][CLΣ] 6 [Σ+] 7 [Σ+] 8 [Σ+] 9 [Σ+] 11 [Σ+] 12 [Σ+] 14 [Σ+] 15 [Σ+] [IQR]

**B3:** Q1 = 22.5, Med = 30, Q3 = 42.5, IQR = 20
Setup: Q1/Q3/IQR(30, 15, 45, 25, 35...)
SC-6: [g][CLΣ] 30 [Σ+] 15 [Σ+] 45 [Σ+] 25 [Σ+] 35 [Σ+] 20 [Σ+] 50 [Σ+] 40 [Σ+] 30 [Σ+] [IQR]

**B4:** Range = 28 ft; IQR = 13 ft
Setup: 31 - 18
SC-6: 40 [ENTER] 12 [-] 31 [ENTER] 18 [-]

**B5:** 75% = 45 trees
Setup: 75% of 60
SC-6: 60 [ENTER] 75 [%]

**B6:** 25% of 60 = 15 trees
Setup: 25% of 60
SC-6: 60 [ENTER] 25 [%]

**B7:** Class A (smaller IQR means less spread in the middle 50%)
Setup: 19 - 8
SC-6: 19 [ENTER] 8 [-]

**B8:** 30 + 15 = 45
Setup: 30 + (1.5 × 10)
SC-6: 30 [ENTER] 1.5 [ENTER] 10 [×] [+]

**B9:** Min=18, Q1=22, Med=27, Q3=34, Max=40; IQR = 12
Setup: Q1/Q3/IQR(18, 22, 25, 29, 34...)
SC-6: [g][CLΣ] 18 [Σ+] 22 [Σ+] 25 [Σ+] 29 [Σ+] 34 [Σ+] 40 [Σ+] [IQR]

**B10:** Range = 90, IQR = 8 (IQR is not distorted by the outlier 100)
Setup: Q1/Q3/IQR(10, 12, 14, 16, 18...)
SC-6: [g][CLΣ] 10 [Σ+] 12 [Σ+] 14 [Σ+] 16 [Σ+] 18 [Σ+] 20 [Σ+] 100 [Σ+] [IQR]

## SEC 9.4 (pp.92-93): MEAN ABSOLUTE DEVIATION (MAD) [COUNT-01]

Methods: W1: 3-Step MAD Distance Table | W2: Number-Line Deviation Balance

### PRACTICE A (p.92): KEY IDEAS (#1-#10)

ExA: Mean = 14, MAD = 2.4

**A1:** Mean = 156 cm, MAD = 4.0 cm
Setup: MAD(150, 162, 156)
SC-6: 150 [Σ+] 162 [Σ+] 156 [Σ+] [g][MAD]

**A2:** Mean = 14, MAD = 2.4
Setup: Sort/Load 5 values -> Mean/MAD
SC-6: Load 5 nums -> [g][x-bar], [g][MAD]

**A3:** Mean = 20, MAD = 6.0
Setup: Sort/Load 5 values -> Mean/MAD
SC-6: Load 5 nums -> [g][x-bar], [g][MAD]

**A4:** Mean = 7, MAD = 0 (Zero spread)
Setup: All distances from 7 are 0
SC-6: Mental / Table Check

**A5:** Mean = 28, MAD = 4.8
Setup: Sort/Load 5 values -> Mean/MAD
SC-6: Load 5 nums -> [g][x-bar], [g][MAD]

**A6:** Mean = 13.0 s, MAD = 0.40 s
Setup: Sort/Load 6 values -> Mean/MAD
SC-6: Load 6 decimals -> [g][x-bar], [g][MAD]

**A7:** Mean = 88, MAD = 3.6
Setup: Sort/Load 5 values -> Mean/MAD
SC-6: Load 5 nums -> [g][x-bar], [g][MAD]

**A8:** Team B (a larger MAD indicates greater variation from the me..
Setup: Compare MADs
SC-6: Mental / Table Check

**A9:** Mean = 10; deviations: 4, 0, 1, 3; MAD = 2.0
Setup: Mean=10; dists 4+0+1+3=8 ÷ 4
SC-6: Mental / Table Check

**A10:** Mean = 9, Med = 8, Mode = 8, Range = 9, IQR = 5.5, MAD = 2.4
Setup: 1 entry -> 6 stats!
SC-6: Mental / Table Check

### PRACTICE B (p.93): REAL LIFE (#1-#10)

ExB: Mean = 13.0 s; MAD = 0.40 s

**B1:** Mean = 210°C; MAD = 3.0°C
Setup: MAD(206, 208, 212, 214)
SC-6: [g][CLΣ] 206 [Σ+] 208 [Σ+] 212 [Σ+] 214 [Σ+] [g][MAD]

**B2:** Mean = 10; MAD = 1.6 shots
Setup: Mean/MAD(7, 9, 10, 11, 13)
SC-6: [g][CLΣ] 7 [Σ+] 9 [Σ+] 10 [Σ+] 11 [Σ+] 13 [Σ+] [g][x-bar] [g][MAD]

**B3:** Mean = 50 g, MAD = 0 g (no deviation from the mean)
Setup: 0
SC-6: 0 [ENTER]

**B4:** Mean = 9 s, MAD = 2.0 s; Raven A is more consistent
Setup: MAD(6, 8, 10, 12)
SC-6: [g][CLΣ] 6 [Σ+] 8 [Σ+] 10 [Σ+] 12 [Σ+] [g][MAD]

**B5:** Mean = 85; MAD = 4.8
Setup: Mean/MAD(76, 82, 85, 88, 94)
SC-6: [g][CLΣ] 76 [Σ+] 82 [Σ+] 85 [Σ+] 88 [Σ+] 94 [Σ+] [g][x-bar] [g][MAD]

**B6:** Mean = 15 m, MAD = 1.0 m
Setup: Mean/MAD(15, 16, 14, 15, 17...)
SC-6: [g][CLΣ] 15 [Σ+] 16 [Σ+] 14 [Σ+] 15 [Σ+] 17 [Σ+] 13 [Σ+] [g][x-bar] [g][MAD]

**B7:** Mean = 85°F; deviations 5, 1, 1, 5; MAD = 3.0°F
Setup: MAD(80, 84, 86, 90)
SC-6: [g][CLΣ] 80 [Σ+] 84 [Σ+] 86 [Σ+] 90 [Σ+] [g][MAD]

**B8:** The signed deviations (-5 + 0 + 5) always add to 0
Setup: -5 + 5
SC-6: 5 [+/-] [ENTER] 5 [+]

**B9:** Mean = 5 min; MAD = 2.4 min
Setup: Mean/MAD(2, 3, 4, 5, 11)
SC-6: [g][CLΣ] 2 [Σ+] 3 [Σ+] 4 [Σ+] 5 [Σ+] 11 [Σ+] [g][x-bar] [g][MAD]

**B10:** Mean=16, Med=15, Mode=15, Rng=8, IQR=5.5, MAD=2.4
Setup: Mean/Med/Mode/Range/Q1/Q3/IQR/MAD(12, 15, 15, 18, 20)
SC-6: [g][CLΣ] 12 [Σ+] 15 [Σ+] 15 [Σ+] 18 [Σ+] 20 [Σ+] [g][x-bar] [f][MED] [g][MODE] [RNG] [IQR]..

# Unit 10: Grade 7 Bridge — Signed Rationals, 2-Step Equations & Probability — Practice A & B Complete Key

**Big Ideas Math Grade 7 Bridge — Chapters 1–3 & Chapter 7 • Manipulative Kits: Ten-Frame Zero-Pair Counters (BASE-10) + Algebra Stage (ALGB-01) + Probability Studio (PROB-01)**

**Target Misconception: Students guess signs when adding/subtracting negatives, get -25 instead of +25 for (-5)² on algebraic calculators, undo ×3 before +7 in 3x + 7 = 28, and confuse Permutations (order matters) with Combinations (order doesn't).**

Hands-On Block Remediation: Use Yellow/Red counters to show removing 4 red debt tokens (10 - (-4)) equals +14. Notice how RPN 5 [+/-] [x²] squares the entire register (-5)² = +25 without parentheses! Unwrap 2-step equations in reverse (untie ribbon ±b first, open box ×a second), and finish with [DICE], [f] [nCr], [g] [nPr], and [f] [?].



**QUEST-01 Double-Points Scoring: 1 '+10' token = Textbook Method 1 (1st Way); stack a 2nd '+10' token (+20 pts) = Textbook Method 2 (2nd Way)!**

## SEC 10.1 (pp.95-96): ADDING AND SUBTRACTING RATIONAL NUMBERS [BASE-10]

Methods: W1: Zero-Pair Counters | W2: Number-Line Jumps | W3: Absolute Value Sign Rules & Add-the-Opposite

### PRACTICE A (p.95): KEY IDEAS (#1-#10)

ExA: -8 + 5 = -3; -6 + (-4) = -10

**A1:** (a) -6; (b) -24
Setup: -15 + 9; -9
SC-6: 15 [+/-] [ENTER] 9 [+] / 9 [+/-] [+]

**A2:** -9
Setup: 14 + -23
SC-6: 14 [ENTER] 23 [+/-] [+]

**A3:** (a) 30; (b) -4
Setup: 12 - -18; -12
SC-6: 12 [ENTER] 18 [+/-] [-] / 12 [+/-]...

**A4:** -25
Setup: -9 - 16
SC-6: 9 [+/-] [ENTER] 16 [-]

**A5:** -14.2°C
Setup: (0 - 7.8) - 6.4
SC-6: 0 [ENTER] 7.8 [-] 6.4 [-]

**A6:** -4.8
Setup: 6.4 - 11.2
SC-6: 6.4 [ENTER] 11.2 [-]

**A7:** +2.7
Setup: -7.5 - -10.2
SC-6: 7.5 [+/-] [ENTER] 10.2 [+/-] [-]

**A8:** -1/8
Setup: -3/4 - -5/8
SC-6: [.] 3 [.] 4 [+/-] [ENTER] [.] 5 [.] 8 [+/-] [-]

**A9:** 36°F
Setup: 14 - -22
SC-6: 14 [ENTER] 22 [+/-] [-]

**A10:** 2 7/8 (2.875)
Setup: (-2.5 + 4.25) + 1.125
SC-6: 2.5 [+/-] [ENTER] 4.25 [+] 1.125 [+]

### PRACTICE B (p.96): REAL LIFE (#1-#10)

ExB: 14°C; and -1/8

**B1:** +8°F
Setup: -14 + 22
SC-6: 14 [+/-] [ENTER] 22 [+]

**B2:** -83 meters
Setup: -45 + -38
SC-6: 45 [+/-] [ENTER] 38 [+/-] [+]

**B3:** 4,800 - (-200) = 5,000 meters
Setup: 4800 - -200
SC-6: 4800 [ENTER] 200 [+/-] [-]

**B4:** -15 - (-25) = +10 points
Setup: -15 - -25
SC-6: 15 [+/-] [ENTER] 25 [+/-] [-]

**B5:** +$8.75
Setup: (-12.50 + 30) - 8.75
SC-6: 12.50 [+/-] [ENTER] 30 [+] 8.75 [-]

**B6:** 1/8
Setup: -5/8 + 3/4
SC-6: [.] 5 [.] 8 [+/-] [ENTER] [.] 3 [.] 4 [+]

**B7:** -9/6 = -1 1/2
Setup: -2/3 - 5/6
SC-6: [.] 2 [.] 3 [+/-] [ENTER] [.] 5 [.] 6 [-]

**B8:** -16.0 meters
Setup: (-18.4 + 7.9) - 5.5
SC-6: 18.4 [+/-] [ENTER] 7.9 [+] 5.5 [-]

**B9:** -15.0°C
Setup: -8.5 - 6.5
SC-6: 8.5 [+/-] [ENTER] 6.5 [-]

**B10:** -8
Setup: ((-14 - -20) + -9) - 5
SC-6: 14 [+/-] [ENTER] 20 [+/-] [-] 9 [+/-] [+] 5 [-]

## SEC 10.2 (pp.97-98): MULTIPLYING AND DIVIDING RATIONAL NUMBERS [BASE-10]

Methods: W1: Same/Different Sign Rules | W2: Repeated Addition / Equal Number-Line Jumps & Counter Flips

### PRACTICE A (p.97): KEY IDEAS (#1-#10)

ExA: 6×(-4) = -24, (-6)×(-4) = +24, (-5)² = +25

**A1:** (a) -56; (b) +56
Setup: 7 × -8; -7
SC-6: 7 [ENTER] 8 [+/-] [×] / 7 [+/-]...

**A2:** (a) -5; (b) +5
Setup: -45 ÷ 9; -9
SC-6: 45 [+/-] [ENTER] 9 [÷] / 9 [+/-] [÷]

**A3:** +25 (2 negative factors = positive)
Setup: -5²
SC-6: 5 [+/-] [x²]

**A4:** -8 (3 negative factors = odd count = negative)
Setup: -2³
SC-6: 2 [+/-] [x³]

**A5:** +8.4
Setup: -3.5 × -2.4
SC-6: 3.5 [+/-] [ENTER] 2.4 [+/-] [×]

**A6:** -12
Setup: -14.4 ÷ 1.2
SC-6: 14.4 [+/-] [ENTER] 1.2 [÷]

**A7:** -3/5 (-0.6)
Setup: -2/3 × 9/10
SC-6: [.] 2 [.] 3 [+/-] [ENTER] [.] 9 [.] 10 [×]

**A8:** +2
Setup: -5/6 ÷ -5/12
SC-6: [.] 5 [.] 6 [+/-] [ENTER] [.] 5 [.] 12 [+/-] [÷]

**A9:** -120
Setup: (-4 × -5) × -6
SC-6: 4 [+/-] [ENTER] 5 [+/-] [×] 6 [+/-] [×]

**A10:** -76
Setup: (-18 × 4) - (-36 ÷ -9)
SC-6: 18 [+/-] [ENTER] 4 [×] 36 [+/-] [ENTER] 9 [+/-] [÷] [-]

### PRACTICE B (p.98): REAL LIFE (#1-#10)

ExB: 2 minutes

**B1:** -120 feet
Setup: 8 × -15
SC-6: 8 [ENTER] 15 [+/-] [×]

**B2:** +40 points (same signs give a positive product)
Setup: -4 × -10
SC-6: 4 [+/-] [ENTER] 10 [+/-] [×]

**B3:** -4.5°F per hour
Setup: -27 ÷ 6
SC-6: 27 [+/-] [ENTER] 6 [÷]

**B4:** (-4)³ = +16; (-4)³ = -64
Setup: -4³
SC-6: 4 [+/-] [x³] 4 [+/-] [x³]

**B5:** +2/3
Setup: -3/4 × -8/9
SC-6: [.] 3 [.] 4 [+/-] [ENTER] [.] 8 [.] 9 [+/-] [×]

**B6:** -2
Setup: -7/8 ÷ 7/16
SC-6: [.] 7 [.] 8 [+/-] [ENTER] [.] 7 [.] 16 [÷]

**B7:** -12
Setup: (-2.5 × -1.6) × -3
SC-6: 2.5 [+/-] [ENTER] 1.6 [+/-] [×] 3 [+/-] [×]

**B8:** -3.2 meters per year
Setup: -14.4 ÷ 4.5
SC-6: 14.4 [+/-] [ENTER] 4.5 [÷]

**B9:** +120
Setup: ((-2 × -3) × -4) × -5
SC-6: 2 [+/-] [ENTER] 3 [+/-] [×] 4 [+/-] [×] 5 [+/-] [×]

**B10:** 36 - (-20) + 6 = 62
Setup: (-6² - (4 × -5)) + (-18 ÷ -3)
SC-6: 6 [+/-] [x²] 4 [ENTER] 5 [+/-] [×] [-] 18 [+/-] [ENTER] 3 [+/-]..

## SEC 10.3 (pp.99-100): SOLVING TWO-STEP EQUATIONS (AX + B = C) [ALGB-01]

Methods: W1: Two-Step Inverse Operations | W2: Balance Scale / Bar Model Unwrapping

### PRACTICE A (p.99): KEY IDEAS (#1-#10)

ExA: x = 7 (Step 1: 3x = 21)

**A1:** 4x = 36; x = 9
Setup: (45 - 9) ÷ 4
SC-6: 45 [ENTER] 9 [-] 4 [÷]

**A2:** 5x = 50; x = 10
Setup: (38 + 12) ÷ 5
SC-6: 38 [ENTER] 12 [+] 5 [÷]

**A3:** 6y = 48; y = 8
Setup: (59 - 11) ÷ 6
SC-6: 59 [ENTER] 11 [-] 6 [÷]

**A4:** 8n = 64; n = 8
Setup: (49 + 15) ÷ 8
SC-6: 49 [ENTER] 15 [+] 8 [÷]

**A5:** m / 4 = 9; m = 36
Setup: (15 - 6) × 4
SC-6: 15 [ENTER] 6 [-] 4 [×]

**A6:** k / 3 = 12; k = 36
Setup: (7 + 5) × 3
SC-6: 7 [ENTER] 5 [+] 3 [×]

**A7:** -3x = -12; x = 4
Setup: (2 - 14) ÷ -3
SC-6: 2 [ENTER] 14 [-] 3 [÷/-] [÷]

**A8:** 2.5w = 15; w = 6
Setup: (23 - 8) ÷ 2.5
SC-6: 23 [ENTER] 8 [-] 2.5 [÷]

**A9:** 25h + 35 = 160; h = 5 hours
Setup: (160 - 35) ÷ 25
SC-6: 160 [ENTER] 35 [-] 25 [÷]

**A10:** 15w + 45 = 120; w = 5 weeks
Setup: (120 - 45) ÷ 15
SC-6: 120 [ENTER] 45 [-] 15 [÷]

### PRACTICE B (p.100): REAL LIFE (#1-#10)

ExB: 4 hours

**B1:** 6r + 8 = 38 -> r = 5 laps
Setup: (38 - 8) ÷ 6
SC-6: 38 [ENTER] 8 [-] 6 [÷]

**B2:** 4h + 12 = 48 -> h = 9 hours
Setup: (48 - 12) ÷ 4
SC-6: 48 [ENTER] 12 [-] 4 [÷]

**B3:** 15w + 35 = 140 -> w = 7 weeks
Setup: (140 - 35) ÷ 15
SC-6: 140 [ENTER] 35 [-] 15 [÷]

**B4:** 4b - 10 = 46 -> b = $14 per book
Setup: (46 + 10) ÷ 4
SC-6: 46 [ENTER] 10 [+] 4 [÷]

**B5:** x = 6 (-4·6 + 18 = -6)
Setup: (-6 - 18) ÷ -4
SC-6: 6 [+/-] [ENTER] 18 [-] 4 [+/-] [÷]

**B6:** n = 35
Setup: (16 - 9) × 5
SC-6: 16 [ENTER] 9 [-] 5 [×]

**B7:** 3000 - 40m = 600 -> m = 60 minutes
Setup: (600 - 3000) ÷ -40
SC-6: 600 [ENTER] 3000 [-] 40 [+/-] [÷]

**B8:** x = 24
Setup: (11 + 5) ÷ 2/3
SC-6: 11 [ENTER] 5 [+] [.] 2 [.] 3 [÷]

**B9:** 2w + 36 = 56 -> w = 10 cm
Setup: (56 - 36) ÷ 2
SC-6: 56 [ENTER] 36 [-] 2 [÷]

**B10:** 2.25m + 3.50 = 21.50 -> m = 8 miles
Setup: (21.50 - 3.50) ÷ 2.25
SC-6: 21.50 [ENTER] 3.50 [-] 2.25 [÷]

## SEC 10.4 (pp.101-102): THEORETICAL PROBABILITY, COMBINATIONS & PERMUTATIONS [PROB-01]

Methods: W1: Probability Ratio & Complement Rule | W2: Tree Diagram / Outcome List | W3: Fundamental Counting Principle

### PRACTICE A (p.101): KEY IDEAS (#1-#10)

ExA: P(Blue) = 2/5 = 0.4 = 40%; 5! = 120 ways

**A1:** P(5) = 1/6; P(not 6) = 5/6
Setup: 1/6; and 1 - 1/6 = 5/6
SC-6: Mental / Table Check

**A2:** 0.65 (65% or 13/20)
Setup: 1 - 0.35
SC-6: 1 [ENTER] 0.35 [-]

**A3:** P(double 6s) = 1/36; P(sum 11 or 12) = 3/36 = 1/12
Setup: 1/36; and 3/36 = 1/12
SC-6: Mental / Table Check

**A4:** 24 ways
Setup: 4!
SC-6: 4 [x!]

**A5:** 720 ways
Setup: 6!
SC-6: 6 [x!]

**A6:** 56 teams
Setup: C(8,3)
SC-6: 8 [ENTER] 3 [f][nCr]

**A7:** 336 podium arrangements
Setup: P(8,3)
SC-6: 8 [ENTER] 3 [g][nPr]

**A8:** 1/8 (12.5%)
Setup: 1/2 × 1/2 × 1/2 = 1/8
SC-6: Mental / Table Check

**A9:** 1/3 (6/10 × 5/9 = 30/90)
Setup: 6/10 × 5/9 = 30/90
SC-6: Mental / Table Check

**A10:** 120 combinations vs. 720 permutations (differ by 3! = 6)
Setup: C(10,3); gnPr
SC-6: 10 [ENTER] 3 [f][nCr] / [g][nPr]

### PRACTICE B (p.102): REAL LIFE (#1-#10)

ExB: 720 ordered podiums; 120 unordered 3-runner crews

**B1:** 4/20 = 1/5 = 0.20 = 20%
Setup: Simp(4/20)
SC-6: [.] 4 [.] 20 [g][SIMP] [FDISP]

**B2:** 65% (13/20 = 0.65)
Setup: 100 - 35
SC-6: 100 [ENTER] 35 [-]

**B3:** P(black) = 2/5; P(not white) = 11/15
Setup: Simp(6/15)
SC-6: [.] 6 [.] 15 [g][SIMP]

**B4:** 120 custom skateboards
Setup: (5 × 4) × 6
SC-6: 5 [ENTER] 4 [×] 6 [×]

**B5:** 5! = 120 orders
Setup: 5!
SC-6: 5 [x!]

**B6:** 210 batting orders (permutations)
Setup: P(7,3)
SC-6: 7 [ENTER] 3 [g][nPr]

**B7:** 35 lab groups (combinations)
Setup: C(7,3)
SC-6: 7 [ENTER] 3 [f][nCr]

**B8:** 28 fruit combinations
Setup: C(8,2)
SC-6: 8 [ENTER] 2 [f][nCr]

**B9:** 60 times
Setup: 160 × 3/8
SC-6: 160 [ENTER] [.] 3 [.] 8 [×]

**B10:** 12 outcomes; P(H and 6) = 1/12
Setup: 2 × 6
SC-6: 2 [ENTER] 6 [×]